Online Resource 2 for "Machine-read settlement names did not detectably improve 250 m predictions of mapped river flooding, in a test insensitive to effects near settlements": archive of further tables

This file keeps the tables of the exploratory analyses that neither the paper nor Online Resource 1 cites (the account of the earlier test in Japan is in Text S1 of Online Resource 1), including tables whose quantities were later corrected or superseded; each caption names the amendment that produced it, and the amendments in the public repository record the corrections. Nothing here is needed to read the paper, and no finding rests on these tables alone. Tables numbered Sk are in Online Resource 1. Everything outside the frozen protocol is exploratory (Table S5 of Online Resource 1 gives the planning status of each group of estimates).

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Pre-specified analyses

Table A1: Secondary comparisons under each split: the total increment of names over the recalibrated baseline model (including the regional composition of names) and the flexible model refitted with the name features, for the lexicon and learned score and, on the language-model split, for the language-model reading. The 95 % interval is the fixed-margin interval of the decision table (Sect. 2 of the paper); the fourth column gives the interval from resampling the ratio of the two increments, and the fifth the interval of R minus the same comparison with names permuted among the settlements of each AOI.

Split: comparison	R	95 % interval	Ratio bootstrap, 95 %	real − placebo (95 %)	Verdict	TOST
Location groups: total, M1s→M3s	+0.031	[−0.051, +0.155]	[−0.054, +0.192]	[−0.015, +0.030]	inconclusive	no
Location groups: flexible, M1→M3	+0.038	[−0.058, +0.142]	[−0.072, +0.128]	[−0.003, +0.075]	inconclusive	no
Countries: total, M1s→M3s	−0.037	[−0.149, +0.043]	[−0.275, +0.089]	[−0.052, +0.074]	inconclusive	no
Countries: flexible, M1→M3	+0.040	[−0.047, +0.158]	[−0.079, +0.391]	[−0.143, +0.077]	inconclusive	no
Time: total, M1s→M3s	+0.007	[−0.101, +0.113]	[−0.329, +0.328]	[−0.149, +0.111]	undefined-denominator	—
Time: flexible, M1→M3	−0.128	[−0.513, +0.215]	[−2.026, +1.544]	[−0.168, +0.636]	undefined-denominator	—
Language-model cut-off: total, M1s→M3s	+0.001	[−0.025, +0.028]	[−0.026, +0.087]	[−0.065, +0.022]	equivalent-null	yes
Language-model cut-off: flexible, M1→M3	+0.168	[+0.039, +0.328]	[+0.040, +1.126]	[−0.0004, +0.068]	not-meaning	no
Language-model cut-off: language model, total, M1s→M3rs	+0.027	[+0.0003, +0.056]	[−0.001, +0.195]	[+0.007, +0.045]	positive-small	no
Language-model cut-off: language model, flexible, M1→M3r	+0.052	[−0.023, +0.138]	[−0.034, +0.380]	[−0.161, +0.014]	inconclusive	no

Table A2: Ratio-bootstrap 95 % intervals of R for every comparison and split (post hoc, planned in amendment 2): the two increments are resampled together (cluster bootstrap over location groups, countries for the country hold-out, 4,000 replicates) and their ratio taken in each replicate. The decision table uses fixed-margin intervals instead (Sect. 2 of the paper), because the ratio mixes the uncertainty of the denominator into R. [u]The pre-specified rule leaves R undefined on this split. —: comparison not run on this split.

Comparison	Location groups	Countries	Time[u]	LM cut-off	Events grouped
Names within areas, M1c→M3n (H2)	[−0.016, +0.004]	[−0.008, +0.025]	[−0.089, +0.069]	[−0.014, +0.004]	[+0.001, +0.029]
Names permuted, M1c→M3np	[−0.035, +0.003]	[−0.046, +0.006]	[−0.297, +0.186]	[−0.034, +0.028]	[−0.015, +0.088]
Lexicon only, M1c→M3nx	[−0.019, +0.002]	[−0.002, +0.038]	[−0.113, +0.063]	[−0.015, +0.004]	—
Learned score only, M1c→M3nl	[−0.006, +0.001]	[−0.004, +0.028]	[−0.029, +0.039]	[−0.008, +0.002]	—
Position only, M0c→M0n (H1)	[−0.006, +0.002]	[−0.006, +0.020]	[−0.025, +0.025]	[−0.006, +0.008]	—
Position only, flexible, M0→M0N	[+0.006, +0.252]	[+0.042, +0.287]	[−7.715, +5.075]	[−0.187, +0.011]	—
Total, M1s→M3s	[−0.054, +0.192]	[−0.275, +0.089]	[−0.329, +0.328]	[−0.026, +0.087]	[−0.144, +0.093]
Flexible, M1→M3	[−0.072, +0.128]	[−0.079, +0.391]	[−2.026, +1.544]	[+0.040, +1.126]	—
Language model, M1rc→M3rn (H4)	—	—	—	[−0.034, +0.013]	—
Language model, permuted	—	—	—	[−0.028, +0.003]	—
Language model, total, M1s→M3rs	—	—	—	[−0.001, +0.195]	—
Language model, flexible	—	—	—	[−0.034, +0.380]	—
Names centred within AOIs, M1s→M3d	[−0.012, −0.0004]	[−0.014, +0.0001]	[−0.041, +0.028]	[−0.005, +0.009]	[−0.005, −0.001]

Table A3: Operating characteristics of the decision rule on synthetic labels for each split (the primary split before the freeze; the others post hoc, amendment 2), with 95 % Wilson intervals for the Monte-Carlo error. The simulated effect is δ times the indicator that the nearest name is water-related in the lexicon, calibrated so that $R \approx 0.05$ on the primary split ($\delta = 1.2$, an odds ratio of 3.3); on the other splits the same δ gives the mean R shown, not the smallest effect of interest. Equivalent counts the replicates whose 90 % interval of R lies within ± 0.05, whatever their label. The last column gives, for $\delta = 1.2$, the share of replicates whose 90 % interval of ΔBS lies within the equivalence margin of the real labels on the same split (post hoc, amendment 2): how often an effect of this size would have been called equivalent in our data (—: margin undefined on the time split). Margin ratio and upper ratio (amendment 2, addendum 3): the real-label equivalence margin and the upper end of the observed 95 % interval of ΔBS on the same split, each divided by the mean ΔBS that $\delta = 1.2$ produced there; these ratios ignore that intervals widen under the alternative, so the column before them is the better guide. [n]Includes replicates whose simulated baseline gain was negative (the interval of R is then reversed; Appendix 2.4 of the paper). [a]The effect is injected through the lexicon, so this row is the lexicon test on the language-model split; there is no power analysis for the language-model reading (H4).

Split	δ	Replicates	Equivalent (%)	Positive (%)	Mean R	Labels	Equivalent, real margin (%)	Margin ratio	Upper ratio
Location groups (pre-specified)	0	100	99 [95, 100]	0 [0, 4]	−0.001	equivalent-null 91, inconclusive 1, negative 7, not-meaning 1	—	—	—
	1.2	100	0 [0, 4]	98 [93, 99]	+0.057	inconclusive 1, positive-meaningful 69, positive-small 29, undefined-denominator 1	26 [18, 35]	1.5	0.11
Countries	0	20	95 [76, 99]	0 [0, 16]	−0.004	equivalent-null 16, inconclusive 1, negative 3	—	—	—
	1.2	20	0 [0, 16]	65 [43, 82]	+0.081	not-meaning 4, positive-meaningful 12, positive-small 1, undefined-denominator 3	0 [0, 16]	1.1	0.25
Time	0	20	100 [84, 100]	0 [0, 16]	+0.004	equivalent-null 12, negative 1, undefined-denominator 7	—	—	—
	1.2	20	65 [43, 82]	5 [1, 24]	+0.017	equivalent-null 10, inconclusive 2, positive-negligible 1, undefined-denominator 7	—	—	0.99
Language-model cut-off, lexicon effect[a]	0	20	100 [84, 100]	0 [0, 16]	+0.0002	equivalent-null 18, negative 1, undefined-denominator 1	—	—	—
	1.2	20	85 [64, 95]	15 [5, 36]	+0.013	equivalent-null 13, inconclusive 2, positive-negligible 3, undefined-denominator 2	100 [84, 100][n]	4.2	0.18

Table A4: Descriptive strata for the primary comparison: the H5 strata of the protocol (first block, no intervals as pre-specified; built-up cover and continent are also strata of the protocol and are shown with post hoc intervals) and the post hoc strata by the first language of the country of the nearest settlement (languages with at least 20,000 cells), with 90 % intervals from the cluster bootstrap over location groups within the stratum. —: the baseline gain within the stratum is not positive, so R is undefined. R within a stratum is the names increment in the stratum divided by the baseline gain in the same stratum; small strata have small and unstable baseline gains, hence wide intervals. Language strata are the first language of the country in GeoNames (for example ur for Pakistan); in parentheses, the blocks of the lexicon that read the names of the analysed countries of the stratum (fa and hi for Pakistan). Strata with few location groups have intervals from few resamples, and the fa stratum, three location groups with a large baseline gain of their own, has a small R and a narrow interval.

Stratum	Cells	Flooded (%)	R within stratum	90 % interval
Nearest name in the lexicon	145,456	4.2	−0.007	—
Nearest name not in the lexicon	1,362,030	5.7	−0.003	—
Nearest name water-related	89,404	4.4	−0.004	—
Nearest name not water-related	1,418,082	5.6	−0.003	—
Slope at or below the median	753,457	10.3	−0.004	—
Slope above the median	754,029	0.8	−0.001	—
Built-up cover above 10 %	100,180	1.3	+0.002	[−0.002, +0.007]
Built-up cover at most 10 %	1,407,306	5.8	−0.004	[−0.011, +0.003]
Africa	144,265	7.5	−0.002	[−0.006, +0.003]
Asia	196,090	15.7	−0.010	[−0.028, +0.003]
Europe	824,854	2.3	−0.001	[−0.015, +0.012]
North America	85,031	4.5	−0.001	[−0.008, +0.005]
Oceania	150,806	8.8	+0.006	[−0.009, +0.027]
South America	106,440	5.6	−0.005	[−0.010, +0.0005]
Language: en (en, fr, sw)	403,172	6.3	+0.002	[−0.006, +0.011]
Language: es (es, nah, qu)	169,326	4.4	+0.0001	[−0.007, +0.008]
Language: fr (fr, mg, ng)	122,776	2.8	−0.007	[−0.017, −0.001]
Language: it (de, it)	86,262	1.1	—	—
Language: de (de)	83,889	4.3	−0.005	[−0.054, +0.038]
Language: pt (bantu, pt)	65,887	4.1	−0.016	[−0.036, −0.002]
Language: ur (fa, hi)	61,219	31.5	−0.051	[−0.150, +0.018]
Language: ro (ro)	55,634	1.0	—	—
Language: vi (vi)	48,556	5.1	−0.003	[−0.017, +0.009]
Language: sv (sv)	43,174	0.9	—	—
Language: el (el)	35,780	5.1	−0.006	[−0.016, +0.005]
Language: no (no)	33,813	1.5	+0.007	[−0.044, +0.068]
Language: sq (sq)	28,390	2.6	+0.005	[−0.095, +0.120]
Language: sr (sh)	28,164	2.2	+0.223	[−0.060, +0.898]
Language: fa (fa)	23,990	7.8	+0.0006	[+0.0000, +0.0012]
Language: lv (lv)	20,000	2.1	+0.004	[−0.003, +0.010]

Table A5: Secondary metrics on the primary split (pooled over held-out folds).

Model	Brier	Log loss	PR-AUC	Top-10 % capture (%)
M1 (baseline model)	0.04374	0.15218	0.327	51.0
M3 (refitted with names)	0.04346	0.15161	0.332	51.3
M1s (recalibrated M1)	0.04337	0.15159	0.325	50.9
M3s (+ names)	0.04314	0.15105	0.334	51.7
M1c (+ regional composition)	0.04309	0.15104	0.348	52.9
M3n (+ names within areas)	0.04312	0.15113	0.348	52.9

Table A6: The confirmatory analysis with the frozen code (lexicon not applied to Norwegian names) and with the corrected reading (amendment 4): R of the within-area comparison, 95 % interval and verdict.

Split	R (frozen)	95 %	Verdict	R (corrected)	95 %	Verdict
location groups held out (primary)	−0.003	[−0.013, +0.004]	equivalent-null	−0.004	[−0.015, +0.003]	equivalent-null
countries held out	+0.001	[−0.005, +0.011]	equivalent-null	+0.0005	[−0.003, +0.005]	equivalent-null
time held out	−0.001	[−0.027, +0.023]	undefined-denominator	−0.013	[−0.039, +0.008]	undefined-denominator
after the language model's cut-off	−0.002	[−0.007, +0.002]	equivalent-null	−0.003	[−0.011, +0.003]	equivalent-null

Table A7: Second-stage coefficients on the standardised scale (post hoc, amendment 5): mean over the ten folds of the primary split, with the smallest and largest fold in parentheses. M3n: the name features; M3np: the same features of names reassigned among settlements; M3d: the features centred within each AOI; M1c: the regional composition (AOI means). The reading of Norwegian names is the corrected one (amendment 4). The indicator that the lexicon covers a language of the country is one for every analysed cell with the real names, so its standardised coefficient is zero and the second stage has effectively fourteen name features.

Feature	M3n	M3np (placebo)	M3d	M1c (composition)
logit of the terrain prediction	+1.925 (+1.769, +2.088)	+1.924 (+1.778, +2.085)	+1.969 (+1.883, +2.092)	+1.924 (+1.769, +2.087)
water, nearest name	+0.006 (−0.009, +0.015)	+0.051 (−0.001, +0.108)	+0.003 (−0.003, +0.011)	−0.015 (−0.242, +0.076)
wetland, nearest name	+0.001 (−0.007, +0.006)	−0.023 (−0.030, −0.017)	+0.001 (−0.004, +0.005)	−0.058 (−0.164, +0.083)
riverine, nearest name	−0.003 (−0.012, +0.001)	+0.027 (−0.004, +0.040)	−0.002 (−0.010, +0.005)	+0.145 (+0.003, +0.231)
crossing, nearest name	+0.001 (−0.013, +0.009)	−0.018 (−0.031, −0.004)	+0.005 (−0.007, +0.010)	+0.014 (−0.130, +0.156)
lowland, nearest name	−0.004 (−0.022, +0.011)	−0.025 (−0.042, −0.010)	−0.003 (−0.013, +0.008)	−0.260 (−0.365, −0.170)
highland, nearest name	+0.010 (−0.012, +0.027)	−0.033 (−0.051, −0.005)	+0.010 (−0.004, +0.027)	+0.131 (+0.017, +0.382)
water, three nearest (weighted)	−0.036 (−0.061, −0.019)	−0.059 (−0.089, −0.038)	−0.014 (−0.027, −0.000)	−0.271 (−0.453, +0.098)
wetland, three nearest (weighted)	−0.004 (−0.011, +0.004)	−0.020 (−0.043, −0.006)	−0.007 (−0.013, +0.001)	−0.073 (−0.234, +0.083)
riverine, three nearest (weighted)	+0.023 (+0.003, +0.035)	−0.017 (−0.028, −0.004)	+0.025 (+0.001, +0.038)	−0.037 (−0.174, +0.222)
crossing, three nearest (weighted)	−0.029 (−0.041, −0.016)	+0.003 (−0.008, +0.012)	−0.028 (−0.040, −0.015)	−0.033 (−0.138, +0.162)
lowland, three nearest (weighted)	−0.045 (−0.060, −0.025)	−0.045 (−0.082, −0.015)	−0.032 (−0.041, −0.023)	+0.213 (+0.090, +0.324)
highland, three nearest (weighted)	+0.019 (−0.012, +0.031)	−0.047 (−0.073, −0.012)	+0.023 (−0.004, +0.038)	+0.136 (−0.143, +0.234)
water-related, nearest name	+0.013 (+0.006, +0.023)	+0.037 (+0.015, +0.057)	+0.007 (+0.002, +0.013)	−0.123 (−0.195, −0.067)
lexicon covers the language	+0.000 (+0.000, +0.000)	+0.007 (+0.000, +0.011)	+0.000 (+0.000, +0.000)	+0.000 (+0.000, +0.000)
character-model score	+0.010 (−0.008, +0.026)	−0.017 (−0.090, +0.020)	+0.007 (−0.006, +0.019)	+0.224 (+0.175, +0.292)

Table A8: Interval coverage in the pre-specified null simulations (amendment 2), the p values required by the protocol (amendment 6, addendum 1), the spatial correlation of held-out residuals and reference points for the baseline model (primary split). The CSI maps use the observed flooded count of each AOI and are therefore optimistic. TOST p: no bootstrap replicate fell outside the margin, so p is below $1/B$ for B replicates (the value in brackets).

Quantity	Value
Null simulations: coverage of the 90 % / 95 % interval (mean R)	92 % / 97 %
Null simulations: coverage of zero, 90 % / 95 %	84 % / 92 %
Null simulations: between-replicate SD of R (robust SD; outlying replicates); median bootstrap SE of R	0.0045 (0.0021; 13); 0.0021
Null simulations: verdicts	equivalent-null 91, inconclusive 1, negative 7, not-meaning 1
H2: TOST p	$< 1/B$ (0.00025)
H3: one-sided p ($\Delta_0 > \Delta_4$); Spearman ρ and exact one-sided p	0.86; -0.2, 0.39
H4: TOST p; one-sided p for an improvement	$< 1/B$ (0.00025); 0.66
Holm-adjusted p, H3 / H4	1.00 / 1.00
Correlation of held-out M1 residuals within AOIs by distance (km)	0-0.5: 0.50; 0.5-1: 0.29; 1-2: 0.15; 2-5: 0.04; 5-10: 0.00; 10-20: -0.02
CSI pooled over all cells of maps flagging the observed flooded share per AOI: mean HAND (Table S10) / M1 / M3n	0.25 / 0.30 / 0.30
Brier score: constant (training rate) / M0 / M1	0.0523 / 0.0513 / 0.0437

Exploratory checks of the verdict

Table A9: Corrected equivalence test on the ratio (exploratory, amendment 13, item B1). $L_{\pm} = \Delta \mp 0.05D$ with Δ the change in Brier score from names and D the baseline gain, computed on the same resamples; equivalence when the one-sided 95 % upper limit of L_+ lies below zero and the lower limit of L_- above zero. Fieller: the 90 % interval of R from the variances and covariance of the resamples (normal approximation). (a) 4,000 resamples of location groups with the fitted models held fixed; (b) 500 resamples of location groups with the second stage refitted (corrected code); (c) 500 two-stage resamples, countries then activations within each drawn country, with the second stage refitted; (d) (c) plus the variance of $L_{\pm}$ over 13 assignments of location groups to folds, corrected for overlapping training sets (Nadeau–Bengio; normal approximation). The lower block repeats the double resampling of the protocol rule's operating characteristics for the corrected test (a).

Resampling	R	L_+ upper	L_- lower	Fieller 90 %	Equivalent
(a) Confirmatory run, models fixed	−0.003	-2.4×10^{-4}	1.8×10^{-4}	−0.013 to 0.003	yes
(a) Corrected code, models fixed	−0.004	-2.5×10^{-4}	1.7×10^{-4}	−0.015 to 0.004	yes
(a) H4, nested	−0.003	-1.7×10^{-4}	9.8×10^{-5}	−0.028 to 0.010	yes
(a) H4, total	0.027	2.5×10^{-4}	3.8×10^{-4}	0.003 to 0.161	no
(a) Countries held out	0.001	-6.5×10^{-5}	9.2×10^{-5}	−0.008 to 0.022	yes
(b) Location groups, second stage refitted	−0.004	-2.1×10^{-4}	1.7×10^{-4}	−0.025 to 0.016	yes
(c) Activations within countries, refitted	−0.004	-1.9×10^{-4}	1.1×10^{-4}	−0.042 to 0.031	yes
(d) As (c), with fold-assignment variance	—	-9.6×10^{-5}	3.9×10^{-5}	−0.042 to 0.031	yes
Share of samples declared equivalent with the true R set to −0.05 / 0 / +0.05 (models fixed; Sect. 3.2 of the paper)					
protocol's rule, uniform shift	32 % / 99.9 % / 31 %				
corrected test (a), uniform shift	3.1 % / 98.6 % / 2.6 %				
protocol's rule, shift proportional to the baseline gain	0.0 % / 99.9 % / 0.0 %				
corrected test (a), shift proportional to the baseline gain	15 % / 98.6 % / 2.7 %				
Other measures: the change in Brier units and the increment in Brier skill over a constant forecast (R over a constant forecast, with one definition, in Table A10)					
Primary, confirmatory run	—	ΔBS -2.6×10^{-5}		skill -5.1×10^{-4}	
Primary, corrected code	—	ΔBS -3.1×10^{-5}		skill -5.8×10^{-4}	
H4, nested	—	ΔBS -3.8×10^{-5}		skill -7.3×10^{-4}	
H4, total	—	ΔBS 3.0×10^{-4}		skill 5.7×10^{-3}	

Table A10: The equivalence test on the ratio (exploratory, amendment 13, item B1, reported again under amendment 14, item B1). $L_{\pm} = \Delta \mp 0.05D$, with Δ the change in Brier score from names and D the baseline gain, computed on the same resamples. Each limit is given twice: the 95 % (upper) or 5 % (lower) percentile of the resamples, as the corrected test was planned, and the normal approximation $\hat{L} \pm 1.645\,\mathrm{sd}$ from the variance of the resamples; the two differ where the resampling distribution has heavy tails. TOST p: the larger of the two one-sided normal-approximation p values. Fieller: the 90 % interval of R from the variances and covariance of the resamples; its upper end is the one-sided 95 % upper limit of R. Injected odds ratio: the effect on all ground nearest a water-related name whose mean gain reaches the upper limit, range over the four no-effect references (estimator scale in brackets). (a) 4,000 resamples; (b), (c) and the pigeonhole row 500 resamples with the second stage refitted; (d) adds the Nadeau–Bengio variance over 13 fold assignments to (c), so only the normal approximation is defined. Amendment 11 saved only the percentile interval of the ratio. Monte Carlo standard deviation of the percentile upper limit of L_+ over 1,000 resamples of the replicates: (a) 3.1×10^{-6}, (b) 1.1×10^{-5}, (c) 1.6×10^{-5}. Climatology (training-side constant) Brier score 0.0523: the margin is 0.72 % of it and the upper limit of (d) 0.45 %.

Resampling	L_+ upper: pct / normal	L_- lower: pct / normal	TOST p	Fieller 90 %	Upper R	Injected OR
Amendment 11: activations × countries, refitted (percentile of the ratio)	—	—	—	−0.055 to 0.036	0.036	2.06–2.25
(a) Location groups, models fixed	-2.5×10^{-4} / -2.3×10^{-4}	1.7×10^{-4} / 1.6×10^{-4}	0.001	−0.015 to 0.004	0.004	1.21–1.51 (1.31–1.75)
(b) Location groups, second stage refitted	-2.1×10^{-4} / -1.9×10^{-4}	1.7×10^{-4} / 1.4×10^{-4}	0.003	−0.025 to 0.016	0.016	1.58–1.82 (1.85–2.20)
(c) Activations within countries, refitted	-1.9×10^{-4} / -9.6×10^{-5}	1.1×10^{-4} / 4.0×10^{-5}	0.03	−0.042 to 0.031	0.031	1.93–2.16 (2.37–2.70)
(d) As (c), with fold-assignment variance	— / -9.6×10^{-5}	— / 3.9×10^{-5}	0.03	−0.042 to 0.031	0.031	1.93–2.16 (2.37–2.70)
Pigeonhole: location groups × activations, refitted	-5.4×10^{-5} / 7.2×10^{-5}	2.1×10^{-5} / -1.7×10^{-4}	0.13	−0.096 to 0.067	0.067	2.67–2.83 (3.45–3.68)
R with the baseline gain measured over a constant forecast (Y3: the Brier score of the constant minus that of the recalibrated baseline model M1s; the only definition used in this Supplement) and the Brier skill increment over the constant						
Primary split, confirmatory run	$R = -0.003$ (−0.009 to 0.002)			skill -5.1×10^{-4}		
H4, nested (constant: flooded share of the test cells)	$R = -0.012$ (−0.057 to 0.020)			skill -7.3×10^{-4}		
H4, total (constant: flooded share of the test cells)	$R = 0.098$ (0.016 to 0.187)			skill 5.7×10^{-3}		

Table A11: Robustness of the corrected test to heavy-tailed cluster contributions (exploratory, amendment 14, items B2 and B3), corrected code, primary split, fitted models held fixed. L_+ upper and L_- lower: one-sided 95 % limits (normal approximation from the jackknife variance; bootstrap-t for the studentized row). Fieller: 90 % interval of R (from the jackknife variance, or from 4,000 location-group resamples for the weighted rows). Operating characteristics: 1,000 outer and 500 inner two-stage resamples (countries, then activations within each drawn country); Monte Carlo standard errors in brackets. The last block repeats the double resampling of location groups (2,000 outer and 1,000 inner resamples). An earlier run with other random draws (amendment 12, item B1) gave the protocol's rule 32 % and 33 % at the two margins. Pigeonhole bootstrap (Table A10): 28 of its 500 refitted resamples have a robust z of the change in Brier score above 4, and without them its standard deviation falls from 2.5×10^{-4} to 1.2×10^{-4}; 77 % of the location groups hold a single activation and 70 % of the activations lie within one location group (described after the result, amendment 14, addendum 4).

Variant	R	L_+ upper	L_- lower	Fieller 90 %
Jackknife over location groups (195)	-0.004	-2.3×10^{-4}	1.6×10^{-4}	-0.015 to 0.004
Jackknife over activations (188)	-0.004	-2.2×10^{-4}	1.6×10^{-4}	-0.015 to 0.004
Studentized bootstrap over location groups (4,000)	—	-2.7×10^{-4}	1.9×10^{-4}	—
Without the two most influential groups (PK, IR)	0.0001	-2.1×10^{-4}	2.2×10^{-4}	-0.004 to 0.005
Equal weight per location group	-0.003	-2.9×10^{-4}	2.3×10^{-4}	-0.011 to 0.003
Equal weight per AOI	-0.001	-2.9×10^{-4}	2.8×10^{-4}	-0.005 to 0.003
Equal weight per activation	0.001	-2.4×10^{-4}	2.5×10^{-4}	-0.003 to 0.005
Effective number of clusters G^* (Carter et al., 2017) for the location-group contributions				
Primary split, corrected code	$G = 195$; $G^* = 1.7$ for the change from names, 2.7 for the baseline gain; five groups hold 88 % of the variance			
Primary split, confirmatory run	$G = 195$; $G^* = 2.3$ for the change from names, 2.7 for the baseline gain; five groups hold 85 % of the variance			
Activations kept whole	$G = 106$; $G^* = 2.2$ for the change from names, 3.4 for the baseline gain; five groups hold 96 % of the variance			
Countries held out	$G = 53$; $G^* = 1.4$ for the change from names, 2.7 for the baseline gain; five groups hold 92 % of the variance			
Share of samples declared equivalent (L rule) or non-superior (upper limit only) with the true R set to -0.05 / 0 / $+0.05$; activations within countries, models fixed				
L rule, uniform shift	1.7 % (0.4) / 91.6 % (0.9) / 1.7 % (0.4)			
upper limit only, uniform shift	100.0 % (0.0) / 99.8 % (0.1) / 1.7 % (0.4)			
L rule, shift proportional to the baseline gain	17.7 % (1.2) / 91.6 % (0.9) / 0.7 % (0.3)			
upper limit only, shift proportional to the baseline gain	100.0 % (0.0) / 99.8 % (0.1) / 0.7 % (0.3)			
The same with location groups resampled (amendment 13, confirmatory run): share declared equivalent at $R = -0.05$ / 0 / $+0.05$				
protocol's rule, uniform shift	32.4 % / 99.9 % / 31.2 %			
L rule, uniform shift	3.1 % / 98.6 % / 2.6 %			
protocol's rule, proportional shift	0.0 % / 99.9 % / 0.0 %			
L rule, proportional shift	14.9 % / 98.6 % / 2.7 %			

Table A12: Decomposition of the names increment over the recalibrated baseline model (exploratory, amendment 14, item B5). $\mathrm{BS(M1s)} - \mathrm{BS(M3n)} = [\mathrm{BS(M1s)} - \mathrm{BS(M1c)}] + [\mathrm{BS(M1c)} - \mathrm{BS(M3n)}]$, each divided by the baseline gain of the split: the regional composition (M1s to M1c), the names given the composition (M1c to M3n, the primary estimand), the names centred within AOIs, which contain no composition (M1s to M3d), and the total (M1s to M3s). Brackets: 90 % Fieller intervals of the corrected test (a), 4,000 location-group resamples (unbounded where the baseline gain is not distinguishable from zero). Confirmatory rows take M3d from the run with centred features on the same cells.

Split	BS(M1s)	BS(M1c)	BS(M3n)	Composition	Names	Centred (M3d)	Total (M3s)
Primary (corrected)	0.04337	0.04305	0.04308	0.043	-0.004	-0.004 (-0.010 to -0.001)	0.031 (-0.060 to 0.143)
Language-model cut-off (corrected)	0.04941	0.04988	0.04991	-0.042	-0.003	-0.0002 (-0.003 to 0.007)	0.003 (-0.017 to 0.079)
Primary (confirmatory)	0.04337	0.04309	0.04312	0.037	-0.003	-0.004 (-0.010 to -0.001)	0.031 (-0.058 to 0.142)
Activations kept whole (confirmatory)	0.04767	0.05176	0.05159	-0.430	0.018	-0.003 (-0.005 to -0.001)	-0.032 (-0.135 to 0.091)
Countries (confirmatory)	0.04789	0.04849	0.04848	-0.085	0.001	-0.003 (-0.020 to -0.001)	-0.037 (-0.257 to 0.099)
Time (confirmatory)	0.04605	0.04677	0.04677	-0.311	-0.001	-0.004 (unbounded)	0.007 (unbounded)
Language-model cut-off (confirmatory)	0.04941	0.04974	0.04976	-0.030	-0.002	-0.0003 (-0.004 to 0.007)	0.001 (-0.018 to 0.073)

Table A13: Cluster runs of amendment 14 (exploratory, items C1 and C3). Observable ground: cells whose 10 m built-up and tree cover sum to 0.5 or more left out of evaluation, the cap drawn from the rest. Language-model reading: the total increment on floods after the model's training cut-off, with and without country indicators in the reference model. R with the 90 % interval of the protocol's rule and its verdict; last column: 90 % Fieller interval of the corrected test (a).

Comparison	Cells	Baseline gain	R (90 %)	Verdict	Corrected (a)
Observable ground only: M1c to M3n	1,303,026	9.58×10^{-3}	0.005 (-0.001 to 0.011)	equivalent-null	-0.002 to 0.016
Language-model reading: M1s to M3rs	268,762	1.12×10^{-2}	0.027 (0.004 to 0.051)	positive-small	0.003 to 0.147
Language-model reading with country indicators: M1k to M3rk	268,762	1.12×10^{-2}	0.019 (-0.005 to 0.053)	inconclusive	-0.044 to 0.038
Brier scores on the cut-off split	M1s 0.0494, M3rs 0.0491, M1k 0.0543, M3rk 0.0541				

Table A14: The names increment by AOI (exploratory, amendment 14, item B15), corrected code, primary split: change in Brier score from M1c to M3n (positive when names improve) and change in the area under the ROC curve (AOIs with both classes), by continent and flood type; and R with each of the five activations of largest jackknife influence left out (fitted models held fixed). Continents follow the country of the settlements nearest most of an AOI's cells (GeoNames); the paper and Table S1 follow the continent of the EMS activation, and the two differ for two AOIs between Europe and Asia.

AOIs	n	Median ΔBS	Share > 0	Median ΔAUC	Share > 0
All AOIs	500	-2.5×10^{-6}	47 %	0.0000	49 %
AF	40	-1.5×10^{-5}	40 %	0.0000	53 %
AS	55	4.8×10^{-5}	58 %	0.0000	51 %
EU	328	-1.8×10^{-6}	48 %	0.0000	49 %
NA	16	-1.2×10^{-5}	25 %	−0.0005	31 %
OC	45	1.9×10^{-6}	51 %	0.0000	49 %
SA	16	-1.9×10^{-5}	31 %	−0.0001	44 %
Typed	346	-1.7×10^{-6}	49 %	0.0000	48 %
Untyped (inland, counted as riverine)	154	-3.8×10^{-6}	44 %	0.0000	50 %
R without each of the five most influential activations: EMSR838 0.0002, EMSR097 −0.003, EMSR419 −0.005, EMSR629 −0.005, EMSR567 −0.005					

Table A15: Direction of the exploratory interval estimates that exclude the no-effect value, in the final census (exploratory, amendment 14, item B10). Favouring names: more flooding (or lower, wetter sites) near water-related names, less near highland names, or a positive R. Rows that the rule could not classify (mostly estimates without a class in their key) and rows without a direction are counted together.

Family	Exclude null	Favour names	Against names	Unclassified
Flooding (labels or Brier score)	79	10	57	12
Settlement sites (no flood labels)	98	66	2	30
Other terms (bands, base logit)	125	0	0	125

Table A16: Models of the post hoc analyses. All use the second stage of the primary comparison (Table 1 of the paper); the amendment is the one that planned the model.

Code	Information	Amendment
M3d	logit(M1) + name features centred within each AOI, without composition terms	2
M1co, M3o	as M1c and M3n with the HAND of the nearest settlement's site in place of the names	5
M1cL, M3oL	as M1c and M3n with the flooded share within 500 m of the nearest settlement in the same event (the cell itself left out)	6
M1cG, M3oG	as M1c and M3n with recurrent surface water within 500 m of the nearest settlement (occurrence and seasonality, 1984–2021)	6
M1cS, M3nS, M3nSp	as M1c, M3n and M3np with the strict reading (no springs or wells)	6
T0^{-}	T0 without river distances and without the water and wetland classes of land cover (rung of the ladder)	6
M1cEc, M3oEc	as M1c and M3n with the share of other EMS events (more than 30 days apart) in which the cell itself flooded, and whether there is such an event	7
M1cE, M3oE	the same share averaged over the cells within 500 m of each settlement and assigned through the nearest settlement	7
M1cF, M3nF, M3nFp	as M1c, M3n and M3np with the floodplain reading (riverine and wetland classes)	7
M1 + c(q), M1 + c(q) + q	reference and names models of a reassignment q, with the regional composition recomputed from the reassigned names	7
M1cW, M3oW	as M1c and M3n with recurrent surface water of the cell itself (occurrence, seasonality and maximum extent, 1984–2021, averaged over the cell)	8

Table A17: Post hoc checks of the within-area comparison (M1c to M3n or its analogue): R with its 90 % cluster-bootstrap interval and the verdict of the protocol's decision table. Amendment: the amendment that planned the check; code: the frozen code of the confirmatory run or the code with the corrected reading of Norwegian names. The measure of the baseline gain is that of each run (for the European AOIs, that of these AOIs).

Check	Amendment	Code	R	90 % interval	Verdict
Flood events held out whole	2	frozen	+0.018	[+0.001, +0.039]	not-meaning
Flood events held out whole	7	corrected	+0.004	[−0.004, +0.014]	equivalent-null
Flood traces counted as not flooded	5	corrected	−0.001	[−0.007, +0.004]	equivalent-null
Flooded share over land only	5	corrected	+0.003	[−0.002, +0.009]	equivalent-null
Partly flooded cells excluded	7	corrected	−0.004	[−0.010, +0.001]	equivalent-null
Every AOI weighted equally	6	frozen	−0.0003	[−0.004, +0.004]	—
European AOIs alone, own baseline gain	7	corrected	−0.0005	[−0.013, +0.013]	equivalent-null
Without the Americas and Oceania	5	corrected	−0.006	[−0.015, +0.002]	—
Strict reading	6	corrected	−0.003	[−0.006, +0.0004]	equivalent-null
Gazetteer cleaned of city sections and duplicates	6	corrected	−0.003	[−0.007, +0.0004]	equivalent-null
Without AOIs lacking a hydrography layer	6	corrected	−0.002	[−0.006, +0.004]	equivalent-null
Without AOIs lacking a flood type	6	corrected	−0.009	[−0.019, −0.001]	negative
Language-model reading, primary split	6	corrected	−0.001	[−0.008, +0.005]	equivalent-null
Floodplain reading (riverine and wetland classes)	7	corrected	−0.002	[−0.004, +0.0003]	equivalent-null
Cleaned lexicon (two-letter affixes and English suffixes removed)	8	corrected	−0.004	[−0.009, +0.001]	equivalent-null
GFM labels, existing predictions rescored	3	frozen	−0.003	[−0.020, +0.011]	equivalent-null

Table A18: The primary comparison in subsets of cells and under variants of the fold assignment (post hoc). For subsets, R is the names increment in the subset divided by the baseline gain in the same subset (as for the H5 strata); the 90 % interval comes from the cluster bootstrap over location groups restricted to the subset, and the last column gives the equivalence test alone. For variants, the last column gives the label of the pre-specified decision table and the equivalence test; the run with one seed instead of five is the pre-specified sensitivity analysis 5 (protocol §7-5). Built-up and tree cover removed: cells with ESA CCI built-up cover above 10 % or tree cover above 50 %. Distance bands up to 500 m, 0.5–1 km and beyond 1 km, the classes, the sensors, the fold reassignments and the activation grouping were planned in amendment 2 before they were run. The baseline gain within a class differs between classes by up to about seventeen-fold, so the verdicts by class depend strongly on their own denominators; Table A42 compares classes on the Brier scale. [a]Not planned (added while the analysis was written, or a subset of such an analysis). Values of R for subsets with a small baseline gain (for example single classes of names) are unstable; the increments by class on the Brier scale are in the table of increments by class.

Subset or variant	Cells	Units	Baseline ΔBS	R	90 % interval	Verdict
Distance from the cell to its nearest settlement						
up to 500 m	160,239	195	6.48×10^{-3}	−0.003	[−0.011, +0.004]	equivalent
up to 250 m[a]	44,030	195	6.65×10^{-3}	−0.001	[−0.008, +0.005]	equivalent
250–500 m[a]	116,209	195	6.41×10^{-3}	−0.004	[−0.012, +0.004]	equivalent
0.5–1 km	312,024	195	6.67×10^{-3}	−0.002	[−0.011, +0.006]	equivalent
more than 1 km	1,035,223	195	7.99×10^{-3}	−0.004	[−0.011, +0.002]	equivalent
up to 500 m, nearest name water-related[a]	10,082	134	5.84×10^{-3}	+0.060	[−0.036, +0.162]	not equivalent
up to 500 m, built-up and tree cover removed[a]	111,426	193	8.33×10^{-3}	−0.005	[−0.012, +0.003]	equivalent
up to 500 m, water-related, built-up and tree cover removed[a]	6,945	130	7.70×10^{-3}	+0.051	[−0.050, +0.155]	not equivalent
Class of the nearest name						
water-related (any)	89,404	146	1.25×10^{-2}	−0.004	[−0.023, +0.016]	equivalent
water	35,814	118	2.65×10^{-2}	+0.011	[−0.001, +0.025]	equivalent
wetland	2,857	51	1.54×10^{-3}	−0.499	[−1.262, −0.0002]	not equivalent
riverine	13,841	85	5.81×10^{-3}	−0.244	[−0.509, +0.004]	not equivalent
crossing	13,918	83	1.97×10^{-3}	+0.106	[−0.135, +0.395]	not equivalent
lowland	24,552	101	2.71×10^{-3}	+0.074	[−0.090, +0.265]	not equivalent
highland	58,021	130	8.62×10^{-3}	−0.013	[−0.038, +0.013]	equivalent
Post-event imagery of the delineation						
radar only	839,258	122	5.41×10^{-3}	−0.006	[−0.024, +0.008]	equivalent
optical only	174,932	37	2.19×10^{-2}	+0.0003	[−0.003, +0.004]	equivalent
not recorded (mostly schema generations 0–1)	493,296	65	6.10×10^{-3}	−0.004	[−0.012, +0.004]	equivalent
Fold assignment, grouping and seeds (frozen code; verdict of the decision table)						
pre-specified folds	1,507,486	195	7.55×10^{-3}	−0.003	[−0.011, +0.003]	equivalent-null (TOST holds)
folds reassigned (seed 1)	1,507,486	195	7.50×10^{-3}	−0.005	[−0.012, +0.0003]	equivalent-null (TOST holds)
folds reassigned (seed 2)	1,507,486	195	7.04×10^{-3}	+0.005	[−0.001, +0.013]	equivalent-null (TOST holds)
AOIs of the same activation grouped	1,507,486	106	9.53×10^{-3}	+0.018	[+0.001, +0.039]	not-meaning (TOST holds)
one seed instead of five	1,507,486	195	5.69×10^{-3}	−0.004	[−0.011, +0.002]	equivalent-null (TOST holds)

Table A19: The verdict in more detail (exploratory, amendment 8). Achieved bound: the smallest symmetric margin within which the 90 % interval of R lies. Location groups needed: the multiple of the present number of groups at which the 90 % interval would lie within ±0.05 (equivalence) or the lower 95 % limit would exceed zero (positive), if the half-width shrinks with the square root of the number of groups and the point estimate stays; — where the point estimate lies outside the margins. These multiples are power calculations at the observed point estimate and add nothing to the intervals; the paper no longer uses them. Permutations: 200 reassignments of names among the settlements of each AOI with the regional composition rebuilt. Refitted interval: 500 bootstrap replicates of location groups in which the second stage of both models is refitted with the replicate's weights, against the same replicates with the fitted models held fixed. Effect at the margin: an odds ratio of 2.34 for all ground nearest a water-related name injected into the labels, every model refitted, and each replicate judged by the protocol's decision table without the placebo condition.

Quantity	Value
Achieved equivalence bound, frozen code (verdict): R; Brier units	0.011; 8.3×10^{-5}
Achieved equivalence bound, corrected reading: R; Brier units	0.013; 9.5×10^{-5}
largest over five measures of the baseline gain and two bootstraps (frozen code)	0.046 (two-way, Y1)
Location groups needed, total increment: equivalence; positive (multiple of 195)	20.2; 11.3
Location groups needed, flexible models refitted: equivalence; positive (multiple of 195)	45.7; 7.0
Location groups needed, within 500 m of water-related names (not planned): equivalence; positive (multiple of 134)	—; 3.8
Permutations with rebuilt composition: one-sided p; mean and 95th percentile of R	0.72; −0.002, 0.004
label with the permutation distribution as the placebo	unchanged (equivalent-null)
Corrected run, second stage refitted in each replicate: 90 % (95 %) interval of R	−0.017 to 0.013 (−0.024 to 0.019)
Corrected run, same replicates, models fixed: 90 % (95 %) interval of R	−0.013 to 0.003 (−0.014 to 0.004)
standard deviation of R, refitted over fixed	2.31
Effect at the margin injected, every model refitted: replicates judged equivalent (Wilson 95 %)	0 of 40 (0.0 %–8.8 %)
labels; mean R	positive-small 40; 0.040

Table A20: The skill of the held-out predictions of the corrected run with and without the names, decomposed (exploratory; amendment 9). Murphy decomposition (Murphy, 1973) over ten bins of predicted probability (the binned form omits the within-bin variance and covariance terms, so reliability minus resolution plus uncertainty does not reproduce the Brier score exactly); within AOIs: each model's predictions shifted on the logit scale so that its mean in each AOI equals the observed flooded fraction (479 AOIs with both flooded and dry cells), and the comparison and measure of the baseline gain recomputed; maps: in each AOI, the cells with the highest predicted probability (for HAND, the lowest HAND, ties broken by mean HAND and then at random) flagged as flooded, as many as were observed flooded (479 AOIs). Differences of the decomposition and of the critical success index: M3n minus M1c; the names raised the index in 16 % of the AOIs and lowered it in 23 %. M1c: without names; M3n: with names; M1 ranks cells within AOIs as M1c does. Intervals: 95 % from a cluster bootstrap over location groups (4,000 replicates) unless stated.

Quantity	M1c	M3n	Difference or reference
Decomposition of the Brier score (Murphy, 1973)			
Brier score	0.04305	0.04308	3.1×10^{-5}
Reliability (lower is better)	2.0×10^{-4}	2.1×10^{-4}	8.0×10^{-6} (-5.7×10^{-6} to 3.6×10^{-5})
Resolution (higher is better)	8.0×10^{-3}	8.0×10^{-3}	-1.3×10^{-5} (-4.2×10^{-5} to 1.3×10^{-5})
Uncertainty	5.2×10^{-2}		
Within AOIs (each AOI's flooded fraction given)			
Baseline gain within AOIs	7.5×10^{-3}		
Names increment BS(M1c) − BS(M3n)			-3.1×10^{-5} (-5.8×10^{-5} to -7.2×10^{-6})
R (90 % interval)			−0.004 [−0.007, −0.001]
Maps flagging the observed number of flooded cells per AOI (median; quartiles)			
Critical success index	0.129 (0.03–0.27)	0.132 (0.03–0.27)	mean HAND: 0.084 (0.00–0.21)
Hit rate	0.229 (0.07–0.42)	0.233 (0.06–0.43)	mean HAND: 0.155
False-alarm ratio	0.771 (0.58–0.93)	0.767 (0.57–0.94)	mean HAND: 0.845
Difference in the critical success index			median 0.000 (0.000 to 0.000)

Table A21: The primary comparison under the robustness runs of amendment 9 (exploratory; corrected code; the primary row uses the frozen subsampling seed 20261025). Cells: evaluated cells; baseline gain: BS(M0) − BS(M1) in units of 10^{-3}; odds ratio: flooding within 500 m of water-related names relative to other names, given the held-out terrain prediction of the run, with a pairs cluster bootstrap 95 % interval (2,000 replicates); not computed for the seeds. Permanent water: cells whose mean JRC GSW occurrence at 25 points is at least 50 % were removed in every AOI (32,914 cells, 2.2 %, in 271 AOIs; 14,527 in AOIs with a hydrography layer and 18,387 in AOIs without one), in addition to the water of each product's hydrography layer. Seeds: the seed that draws cells within location groups under the cap (frozen: 20261025). Bare-earth elevation: the elevation features rebuilt from FABDEM v1-2 (Hawker et al., 2022), and HAND replaced by the heights of each cell above the nearest vertex of a HydroRIVERS river draining at least 100 and 1000 km^2; the location model M0 is unchanged. Its baseline model flagged the observed number of flooded cells per AOI with a median critical success index of 0.12 (with GLO-30 and HAND: 0.13); with the names, the median difference was 0.000. By the rule fixed in amendment 9, these runs do not replace the confirmatory verdict, and an odds ratio whose interval excludes one would remain an uncorroborated description.

Run	Cells	Baseline gain	R	90 % interval	Verdict	Odds ratio (95 %)
Primary (corrected code)	1,507,486	7.55	−0.004	[−0.013, +0.003]	equivalent-null	1.01 (0.77–1.31)
GSW permanent water removed	1,474,572	8.58	−0.004	[−0.009, −0.0002]	equivalent-null	1.04 (0.78–1.38)
Subsampling seed 1	1,507,486	7.48	−0.002	[−0.006, +0.003]	equivalent-null	—
Subsampling seed 2	1,507,486	7.46	+0.0004	[−0.004, +0.005]	equivalent-null	—
Subsampling seed 3	1,507,486	8.12	−0.001	[−0.004, +0.003]	equivalent-null	—
FABDEM terrain (bare earth)	1,507,486	6.55	−0.002	[−0.006, +0.003]	equivalent-null	1.02 (0.78–1.33)

Table A22: Robustness of the primary comparison and alternative estimands (exploratory, amendment 10). R with 90 % and 95 % cluster-bootstrap intervals over location groups; in subsets the baseline gain is measured on the same subset. Sensors are those of the EMS post-event images (Sentinel-1 if any image is Sentinel-1). Part C reruns the second stage on the primary split with the corrected code and applies the decision table of the protocol, without replacing the confirmatory verdict; the JRC maps are estimated from a reanalysis that includes the event years. [a]The logit of each model's held-out prediction is shifted within each AOI so that its mean equals the observed flooded fraction, and AOIs that are all flooded or all dry are left out; the baseline gain is measured in the same way. The placebo rows were added after the GFM result had been seen (addendum 2 of amendment 10), because the decision table requires them for an interval above zero; the last column gives R minus the placebo's R with its 95 % interval. [b]The estimator of Eq. (2) of the paper (AOI fixed effects, distance bands and the logit of the baseline model's held-out prediction) on the cells of each subset, with a 95 % pairs cluster bootstrap over location groups (2,000 replicates); not estimated when fewer than five water-related cells within 500 m flooded. [c]The pairs bootstrap separated in some replicates (standard deviation of the log odds ratio above 3), so the cluster-robust interval is given.

Comparison	R	90 % interval	95 % interval	Verdict or sample
Primary comparison in subsets (corrected code; base models of the full run)				
EMS map from Sentinel-1	−0.018	−0.156 to 0.010	−0.289 to 0.015	160 AOIs, 77 groups
EMS map from other radar	−0.001	−0.008 to 0.019	−0.011 to 0.032	132 AOIs, 70 groups
EMS map from optical images	0.003	−0.002 to 0.026	−0.003 to 0.040	56 AOIs, 37 groups
Images not recorded	−0.007	−0.016 to 0.001	−0.018 to 0.003	152 AOIs, 65 groups
Product schema generation 0	−0.002	−0.054 to 0.019	−0.107 to 0.025	31 AOIs, 12 groups
Product schema generation 1	−0.011	−0.020 to −0.001	−0.023 to 0.001	119 AOIs, 56 groups
Product schema generation 2	0.001	−0.003 to 0.007	−0.006 to 0.009	61 AOIs, 28 groups
Product schema generation 3	−0.004	−0.025 to 0.007	−0.037 to 0.010	289 AOIs, 132 groups
Without the African AOIs	−0.005	−0.020 to 0.004	−0.025 to 0.005	460 AOIs, 177 groups
Odds ratio within 500 m of water-related names in the same subsets[b]				
EMS map from Sentinel-1	0.90		0.61–1.30	45 flooded cells
EMS map from other radar	1.44		0.87–2.42	86 flooded cells
EMS map from optical images	0.57		0.44–0.91	43 flooded cells
EMS map from images not recorded	1.02		0.62–1.60	187 flooded cells
Product schema generation 0	0.93		0.28–3.16 (cluster-robust)[c]	25 flooded cells
Product schema generation 1	1.14		0.55–1.87	122 flooded cells
Product schema generation 2	—		too few flooded cells	4
Product schema generation 3	1.02		0.80–1.36	210 flooded cells
Alternative second stages and references (amendment 10, part C)				
Name features × distance bands (M1cd → M3nd)	−0.008	−0.017 to 0.001	−0.019 to 0.003	equivalent-null
Name features × continents (M1cr → M3nr)	−0.062	−0.108 to −0.024	−0.117 to −0.018	negative
Penalty by grouped inner cross-validation (M1ct → M3nt)	−0.013	−0.032 to 0.005	−0.036 to 0.008	equivalent-null
Country indicators in the reference (M1kc → M3kn)	−0.005	−0.011 to 0.001	−0.012 to 0.003	equivalent-null
Baseline model with JRC flood hazard maps (M1hc → M3hn)	−0.004	−0.007 to −0.001	−0.007 to −0.001	negative
Regional composition after the country indicators (M1k → M1kc), share of the baseline gain	−0.029		−0.233 to 0.170	
Scored within AOIs, each AOI's flooded fraction given[a]				
EMS labels, corrected run (amendment 9, item B4)	−0.004	−0.007 to −0.001	−0.008 to −0.001	negative
reassigned names (placebo); real minus placebo	−0.006		−0.010 to −0.001	0.001 (−0.004 to 0.007)
GFM labels, every model refitted (amendment 10, item B10)	0.011	0.004 to 0.019	0.003 to 0.021	positive-negligible
reassigned names (placebo); real minus placebo	−0.017		−0.033 to −0.005	0.028 (0.016 to 0.044)

Table A23: Brier scores of the held-out predictions of each split (exploratory, amendment 10, item B9). Code: the frozen code of the confirmatory run or the corrected code; on the activation split the baseline gain is Y0 9.53×10^{-3}, Y1 3.94×10^{-3}, Y2 5.12×10^{-3}, Y3 4.75×10^{-3}, Y4 3.93×10^{-4}; Table S7 defines the measures of the baseline gain and the bootstraps. On the activation split the reference model M1c is worse than the recalibrated baseline model M1s. Lower block: R under each measure, with the 90 % interval of the cluster and the two-way bootstrap ([†]: the interval does not lie within ±0.05).

Split	Code	M0	M1	M1s	M1c	M3n
Location groups (primary)	corrected	0.0513	0.0437	0.0434	0.0430	0.0431
Countries	frozen	0.0565	0.0494	0.0479	0.0485	0.0485
Time (from 2024)	frozen	0.0495	0.0472	0.0460	0.0468	0.0468
Language-model cut-off	corrected	0.0611	0.0499	0.0494	0.0499	0.0499
Activations kept whole	corrected	0.0583	0.0488	0.0477	0.0519	0.0518
Activations kept whole	frozen	0.0583	0.0488	0.0477	0.0518	0.0516
R on the activation split (corrected code) under each measure of the baseline gain: 90 % interval, cluster bootstrap and two-way bootstrap						
Y0: pre-specified		0.004	−0.004 to 0.014	−0.007 to 0.021		
Y1: lower 95 % bound		0.009	−0.008 to 0.033	−0.036 to 0.116[†]		
Y2: recalibrated		0.007	−0.007 to 0.025	−0.012 to 0.039		
Y3: from a constant		0.008	−0.007 to 0.027	−0.013 to 0.042		
Y4: stacked		0.091	−0.085 to 0.332[†]	−0.158 to 0.508[†]		

Table A24: Further checks of amendment 11 (exploratory). B1: the change of R on the activation split between the frozen and the corrected code, by location group. B6: a 90 % interval from the corrected variance of Nadeau and Bengio (2003) over the 13 fold assignments (it reflects refitting, not the sampling of test cells). B7: 500 replicates that refit the second stage and recompute the baseline gain under each resampling. B4: the predictions of the corrected run, trained on EMS labels, scored on EMS labels. B5: labels with the flooded fraction measured over the observed area that is not permanent water. B10: the pilots, with synthetic labels calibrated to the real flooded fraction of each AOI, in which names and reassigned names improved the nested comparison alike. C1: on the country split, the names reassigned within each AOI 100 times with the regional composition recomputed from each reassignment (corrected code).

Quantity	Value	Note
Activation split: frozen and corrected code (item B1)		
R, frozen code; corrected code	0.018; 0.004	M0 and M1 identical in the two runs
Change of R; from location groups with Norwegian cells	−0.015; −0.00003	6 groups
unit g00098 of the activation split (PK, 38,140 cells)	−0.010	
unit g00054 of the activation split (IR, 14,076 cells)	−0.002	
unit g00076 of the activation split (PK, 20,000 cells)	−0.001	
Intervals for R on the primary split (items B6 and B7)		
Refitting over 13 fold assignments, Nadeau–Bengio, 90 %	−0.003 to 0.0004	mean −0.001
Second stage refitted, ratio bootstrap, activations × countries, 90 %	−0.055 to 0.036	not within ±0.05
Second stage refitted, ratio bootstrap, location groups, 90 %	−0.020 to 0.008	within ±0.05
Country split: reassignments with the composition recomputed (item C1)		
R of the real names (corrected code); mean of 100 reassignments (2.5–97.5 %)	0.0005; −0.003 (−0.026 to 0.012)	
Percentile of the real names; one-sided p	69th; 0.31	
Domain of the GFM labels: EMS-trained predictions on EMS labels (item B4)		
Baseline gain and R, all cells of the GFM AOIs	1.03×10^{-2} (1.83×10^{-3} to 2.17×10^{-2}); −0.005	545,793 cells
Baseline gain and R, cells that GFM can assess	1.57×10^{-2} (2.48×10^{-3} to 3.44×10^{-2}); −0.006	234,703 cells
Denominator of the flooded fraction (item B5)		
Cells partly unobserved or partly permanent water: all; within 500 m of water-related; of other names	4.2 %; 5.0 %; 4.3 %	
Relabelled over the observed non-permanent-water area: changed to flooded; left out	1,541; 38	
R rescored on these labels (90 %)	−0.004 (−0.016 to 0.003)	
Pilot simulations before the freeze, no injected effect (item B10)		
Pilot 3, rep0000: nested increment R of names; of reassigned names	0.185; 0.153	
Pilot 3, rep0001: nested increment R of names; of reassigned names	0.094; 0.095	

Table A25: Census of the exploratory interval estimates of amendments 2 to 10 (amendment 11, item B8): every estimate with a 95 % interval in the verification outputs v36 to v67, collected by key rules and counted once where the same value appears in several files. Distance-band and logit coefficients, near-minus-far contrasts, intervals that are not finite or inverted, and fits that separate are excluded, and so are the controls (records of surface water, recurrence, oracles and hazard maps). Excludes: the interval excludes the no-effect value (one for odds ratios, zero otherwise). The estimates are strongly dependent (many are variants of the same quantity). The first pass counted the excluded coefficients as name effects; the rules were refined before any count was used. The odds ratios within 500 m of settlements are those of every estimate whose key or record names a near-settlement quantity; the first version of this census flagged them by the key alone and missed 93 of them (an implementation error corrected in addendum 1 of amendment 12). The q values are approximate: they are computed from p values recovered from asymmetric intervals, and the estimates are not independent. The machine-readable list is v69_estimates.csv in the public repository.

Family	Estimates	Excludes	Share
Name effects on flood labels or predictions	641	79	12 %
R	74	8	11 %
difference	78	12	15 %
odds ratio	489	59	12 %
Name effects on terrain or water descriptors (no flood labels)	190	98	52 %
difference	190	98	52 %
Odds ratios within 500 m of settlements	198	24	12 %
Benjamini–Hochberg $q < 0.05$; $q < 0.10$		0; 9	
Benjamini–Yekutieli $q < 0.05$; $q < 0.10$		0; 0	

Table A26: Further checks of amendment 12 (exploratory). B1: shares of resampled samples in which each decision rule declares equivalence when the true R is set by shifting the names increment (uniformly over cells, or in proportion to each location group's baseline gain); the protocol's rule divides the interval of the change in Brier score by the point baseline gain, the ratio rule uses the interval of R. B10: the reference model refitted from the second-stage design of the corrected run. B17: held-out Brier scores of the position model and of the best constant forecast (the test cells' flooded share). B5: the thirteen outlying null replicates of the protocol's simulations. B4, B12, B13: inclusion, GFM images and flood typing. B18: the model of Eq. (2) of the paper with and without the share of built-up land at 10 m (ESA WorldCover 2021). B2: countries of the flooded cells within 500 m of water-related names in the sample of every evaluable cell. B11: the second stage with the penalty chosen by grouped inner cross-validation for the name features and the composition only, on a grid widened until the optimum is interior.

Quantity	Value	Note
Share of resampled samples declared equivalent (item B1)		
True $R = -0.05$, shift uniform over cells: protocol's rule; ratio rule	32 %; 3 %	
True $R = 0$, shift uniform over cells: protocol's rule; ratio rule	100 %; 98 %	
True $R = +0.05$, shift uniform over cells: protocol's rule; ratio rule	33 %; 3 %	
True $R = -0.05$, shift proportional to the baseline gain: protocol's rule; ratio rule	0 %; 15 %	
True $R = 0$, shift proportional to the baseline gain: protocol's rule; ratio rule	100 %; 98 %	
True $R = +0.05$, shift proportional to the baseline gain: protocol's rule; ratio rule	0 %; 3 %	
Reference model on the primary split, corrected code (item B10)		
Brier score: M1s; M1c; M1c without the area mean of the character model's score	0.04337; 0.04305; 0.04382	
Correlation of sibling AOI residuals within 50 km (Fisher 95 %)	-0.74 (-0.92 to -0.29)	12 pairs
Position model against a constant forecast (item B17)		
Brier score of M0; of the best constant, location groups held out	0.0513; 0.0523	better
Brier score of M0; of the best constant, countries held out	0.0565; 0.0523	worse
Brier score of M0; of the best constant, floods from 2024 held out	0.0495; 0.0473	worse
Brier score of M0; of the best constant, floods after the language model's cut-off	0.0611; 0.0525	worse
Brier score of M0; of the best constant, activations kept whole	0.0583; 0.0523	worse
Outlying null simulations of the protocol (item B5)		
Replicates led by location group g00024 (BD); its share of R	5	0.84 to 1.10
Replicates led by location group g00175 (PK); its share of R	8	0.76 to 1.31
Inclusion criteria applied in order (item B4)		
Evaluable cells ≥ 400	47 excluded	952 remaining
Flooded cells ≥ 20	452 excluded	500 remaining
Flooded share ≤ 80 %	0 excluded	500 remaining
Riverine (or untyped inland) share of the flooding ≥ 0.5	0 excluded	500 remaining
A settlement inside the grid	0 excluded	500 remaining
Flood typing (item B13) and the GFM window (item B12)		
AOIs with untyped flooding; counted riverine by the inland rule	150; 154	with the ice-jam AOIs
AOIs with typed flooding of another kind; descriptions	6	Ice jam flood (12449); Landslide (268); None (2)
Flood traces as a share of the flooded area: optical; Sentinel-1; other radar	14 %; 0.0 %; 0.0 %	
Distance of the AOI centroid from the coast: minimum; 10th percentile; median (km)	1.4; 13.8; 79	81 AOIs within 20 km
AOIs with GFM images: with one on an EMS image day; only on adjacent days	240; 77	of 317
Labels near settlements and built-up land at 10 m (item B18)		
Odds ratio, ground 0.5–1 km against within 500 m: without; with the built-up share	1.12 (1.05–1.20); 1.05 (0.98–1.13)	
Odds ratio, ground beyond 1 km against within 500 m: without; with the built-up share	1.29 (1.12–1.48); 1.17 (1.02–1.34)	
Odds ratio, water-related names within 500 m (cluster-robust): without; with the built-up share	1.01 (0.77–1.32); 0.98 (0.75–1.29)	
water-related, pairs bootstrap 95 %: without; with	0.78–1.31; 0.75–1.28	
Every evaluable cell: where the flooded cells within 500 m of water-related names are (item B2)		
Cambodia	1,299 (35 %)	4 AOIs
Myanmar	909 (24 %)	11 AOIs
Bangladesh	384 (10 %)	2 AOIs
Pakistan	264 (7.0 %)	12 AOIs
Portugal	87 (2.3 %)	9 AOIs
Germany	84 (2.2 %)	31 AOIs
Activation EMSR052 (Cambodia)	1,124 (30 %)	3 AOIs
Activation EMSR130 (Myanmar/Burma)	909 (24 %)	11 AOIs
Odds ratio without Pakistan and Bangladesh (pairs 95 %)	0.87 (0.73–1.07)	3,105 flooded cells
Penalty on the name features and the composition only (item B11)		
R; chosen C for M1c and M3n (range over folds)	0.004	10^{-7} to 10^{-4}; 10^{-7} to 10^{-4}; at the edge of the widened grid in 1 of 10 folds

Table A27: Whether areas with more water-related names flood more, within language regions (exploratory; amendment 10, item B13). Regression of the empirical logit of each AOI's flooded fraction, $\log\{(k+0.5)/(n-k+0.5)\}$, on the share of its analysed cells whose nearest name is water-related and on the AOI mean of the baseline model's held-out logit, with activation fixed effects, for the AOIs of countries read with the German, English or French lexicon blocks; AOIs that are the only analysed AOI of their activation drop out with the fixed effects. Standard errors clustered by activation (CR1), 95 % intervals from the t distribution with one degree of freedom fewer than the activations.

Language	AOIs	Activations	Share water-related	Mean logit of M1
German	54	16	−3.58 (−21.02 to +13.85)	+0.47 (+0.10 to +0.84)
English	111	30	−2.27 (−6.39 to +1.86)	+0.57 (+0.17 to +0.97)
French	58	14	−3.62 (−12.16 to +4.92)	+0.79 (−0.06 to +1.64)
All three	212	56	−2.38 (−6.04 to +1.28)	+0.56 (+0.29 to +0.84)

Table A28: Post hoc runs and controls of amendment 6, primary split, corrected code. R is the within-area increment as a fraction of the baseline gain of the same run, with the fixed-margin 90 % interval and the verdict of the decision table. At the T0 rung, the 95 % interval of R is −0.024 to 0.002 with the corrected code against −0.017 to 0.001 with the frozen code (Table 2 of the paper), so a 90 % interval here can be wider than the 95 % interval there. The spread of the location groups' contributions grew by a factor of 1.48, almost all of it from two large AOIs, EMSR097 AOI01 (BD) and EMSR838 AOI05 (PK), whose contributions changed when the models were refitted; without them the factor is 0.87.

Run	R	90 % interval	Verdict	Sample
Flood-label oracle (same event, leaks the outcome)	0.513	0.338 to 0.715	positive-meaningful	500 AOIs
Recurrent surface water (JRC GSW)	0.006	−0.009 to 0.022	equivalent-null	500 AOIs
Recurrent surface water without permanent water	0.004	−0.020 to 0.027	equivalent-null	500 AOIs
Names on T0$^-$ (no river distances, no water land cover)	−0.008	−0.019 to −0.001	negative	500 AOIs
Names on T0 (same run)	−0.008	−0.021 to 0.001	equivalent-null	500 AOIs
Strict reading of the names	−0.003	−0.006 to 0.0004	equivalent-null	500 AOIs
Language-model reading, primary split	−0.001	−0.008 to 0.005	equivalent-null	500 AOIs
Gazetteer without city sections, historical places, duplicates	−0.003	−0.007 to 0.0004	equivalent-null	500 AOIs
Without AOIs lacking a hydrography layer	−0.002	−0.006 to 0.004	equivalent-null	285 AOIs
Without AOIs lacking a flood type	−0.009	−0.019 to −0.001	negative	346 AOIs
Events held out whole: real increment among 200 reassignments	—	percentile 48	—	—
Every AOI weighted equally	−0.0003	−0.004 to 0.004	equivalent	—

What assignment through settlements can carry

Table A29: Injected odds ratio (shape A: all ground nearest a water-related name) whose mean simulated gain reaches each interval's upper limit of R minus the gain expected without an effect, under the four no-effect references (exploratory, amendment 13, item B2); in parentheses, the value on the scale of the estimator of Eq. (2) of the paper. A bound is not a test inversion, and the Monte Carlo ranges of the bound tables are not confidence intervals. References: recomputed composition (amendment 8), no reassignment, centred reassignments, and the reassignments of amendment 5.

Interval	Upper R	Recomputed	None	Centred	Amendment 5
Protocol's interval (models fixed)	0.003	1.25 (1.36)	1.17 (1.24)	1.28 (1.40)	1.47 (1.69)
Location groups, second stage refitted (amendment 8)	0.013	1.57 (1.83)	1.50 (1.73)	1.59 (1.86)	1.76 (2.11)
Location groups, refitted, denominator resampled (amendment 11)	0.008	1.40 (1.59)	1.34 (1.50)	1.43 (1.62)	1.64 (1.94)
Activations × countries, refitted (amendment 11)	0.036	2.11 (2.63)	2.06 (2.55)	2.13 (2.65)	2.25 (2.83)
Spread over fold assignments (amendment 5)	0.004	1.31 (1.45)	1.23 (1.34)	1.33 (1.47)	1.53 (1.77)
Corrected test (a)	0.004	1.29 (1.43)	1.21 (1.31)	1.31 (1.45)	1.51 (1.75)
Corrected test (b)	0.016	1.65 (1.95)	1.58 (1.85)	1.67 (1.97)	1.82 (2.20)
Corrected test (c)	0.031	1.99 (2.45)	1.93 (2.37)	2.00 (2.47)	2.16 (2.70)
Corrected test (d)	0.031	1.99 (2.45)	1.93 (2.37)	2.01 (2.48)	2.16 (2.70)
Effect within 1 km of the settlement (shape BK; amendment 12, item C1)					
Margin	0.050	7.55			
Models fixed		2.13	1.82	2.20	2.92
Second stage refitted		3.19	2.99	3.26	3.98

Table A30: Effects injected into the observed labels for all ground nearest a water-related name, with the base models fixed and with every model refitted on the injected labels (amendment 7; three repetitions with the injections of amendment 6): mean gain in ΔBS over the uninjected run ($\times 10^{-4}$), the same with fixed base models over the ten repetitions of amendment 6, the ratio of the refitted to the fixed gain, the baseline gain (M0 to M1) of the refitted models in the injected labels ($\times 10^{-4}$) and the share of held-out labels changed by the injection (%). Odds ratios at the margin and bounds are not read from this simulation: it has no effect below an odds ratio of 1.65, and a straight line from zero across the convex gain curve would understate them (amendment 8, addendum 9).

Odds ratio	Fixed	Fixed, 10 rep.	Refitted	Ratio	Baseline gain	Labels changed
1.65	1.36	1.37	1.23	0.91	78.9	0.15
2.72	5.12	5.18	4.88	0.95	78.9	0.34
4.48	12.38	12.39	11.66	0.94	78.1	0.55

Table A31: How an effect confined to the ground a name refers to appears in the odds ratio for all ground nearest the name (amendment 10, addendum 1, item 3; arithmetic, no data). φ: share of the nearest ground that the named feature occupies; OR_n: odds ratio of flooding there. Rare-outcome approximation: risk ratio $1+\varphi(\mathrm{OR}_n-1)$ on the ground of truly water-related names, observed through names read with precision $p_1 = 0.76$ and a water-related share $p_0 = 0.18$ among other names, both measured with the codes of the model that compiled the lexicon (Table 5 of the paper and Sect. 3.6 of the paper; p_0 from the names not read as water-related in the sample of Wikipedia derivations).

OR_n	$\varphi = 0.05$	$\varphi = 0.1$	$\varphi = 0.25$	$\varphi = 0.5$
1.5	1.01	1.03	1.07	1.14
2	1.03	1.06	1.14	1.27
3	1.06	1.11	1.27	1.49
5	1.11	1.22	1.49	1.85
Without misreading ($p_1 = 1$, $p_0 = 0$)				
1.5	1.02	1.05	1.12	1.25
2	1.05	1.10	1.25	1.50
3	1.10	1.20	1.50	2.00
5	1.20	1.40	2.00	3.00

Table A32: The margin and the bounds of Sect. 3.3 of the paper as injected (nominal) odds ratios and on the scale of the estimator of Eq. (2) of the paper (exploratory; amendment 11, item B3). The estimator was applied to labels with known injected odds ratios (Table S18); its mean odds ratio, relative to the uninjected labels, is interpolated on the log scale. Beyond the largest injected odds ratio (3) the estimator value is given as a lower limit. The no-effect reference is the mean increment of reassigned names with rebuilt composition. Calibration curve for all ground (nominal → estimator): 1.25 → 1.37, 1.5 → 1.73, 2 → 2.47, 3 → 3.94.

Quantity	Injected odds ratio	Estimator odds ratio
Margin ($R = 0.05$), all ground nearest a water-related name	2.33	2.94
Upper bound, all ground, fitted models held fixed	1.25	1.36
Upper bound, within 500 m, fitted models held fixed	6.25	>4.73
Upper bound, all ground, with the spread over fold assignments	1.31	1.45
Upper bound, within 500 m, with the spread over fold assignments	7.77	>4.73
Upper bound, all ground, with the second stage refitted	1.57	1.83
Upper bound, within 500 m, with the second stage refitted	18.37	>4.73
Upper bound, all ground, flips correlated within each settlement's ground	1.27	1.39
Upper bound, all ground, effects that differ between settlements (mean odds ratio)	1.23	1.33

Table A33: Odds ratios of flooding that the margin and the observed upper limits correspond to under the synthetic labels of the protocol's power analysis (amendment 2): an effect on all ground nearest a water-related name, and on the ground within 500 m of such a settlement. In parentheses the range over Monte Carlo resamples of the simulated curve; >: beyond the largest simulated effect (odds ratio 4.95 for all ground, 20.1 within 500 m). These translations assume the label model of the power analysis; the simulations on the observed labels (Table A65) give lower values.

Target of the translation	All ground nearest the name	Within 500 m
Margin ($R = 0.05$)	4.12 (4.04–4.22)	>20
Upper 90 % limit of the observed increment	1.53 (1.44–1.61)	6.94 (4.12–9.74)
Same, two-way bootstrap	1.72 (1.63–1.83)	9.52 (6.56–12.64)
Upper limit, confirmatory run	1.53	6.94
Upper limit, fold assignment 1	1.19	3.22
Upper limit, fold assignment 2	2.20	>20
Upper limit, location groups formed with a 10 km buffer	1.76	10.14
Upper limit, events held out whole	4.10	>20

Table A34: Positive control without flood labels (amendment 6): within-AOI difference between settlements whose name is in each class and other settlements in the mean occurrence of surface water (percentage points of time), its mean seasonality (months) and the occurrence without permanent water, within 500 m of the settlement point (JRC Global Surface Water 1984–2021); 95 % cluster-bootstrap intervals over location groups.

Names	Occurrence	Seasonality	Occurrence, non-permanent
water-related (any)	+1.05 (+0.34 to +2.00)	+0.12 (+0.04 to +0.23)	+0.60 (+0.13 to +1.25)
strict reading	+1.36 (+0.41 to +2.65)	+0.15 (+0.04 to +0.30)	+0.74 (+0.11 to +1.62)
water	+0.48 (-0.22 to +1.55)	+0.06 (-0.02 to +0.19)	+0.19 (-0.18 to +0.85)
wetland	-0.34 (-1.01 to +0.34)	-0.05 (-0.13 to +0.04)	-0.29 (-0.79 to -0.04)
riverine	+4.04 (+1.88 to +6.01)	+0.43 (+0.20 to +0.67)	+2.56 (+0.76 to +4.17)
crossing	+1.87 (+0.58 to +3.38)	+0.22 (+0.07 to +0.38)	+1.24 (+0.32 to +2.35)
lowland	+0.03 (-0.14 to +0.22)	+0.00 (-0.02 to +0.03)	-0.01 (-0.09 to +0.08)
highland	-0.47 (-0.79 to -0.14)	-0.06 (-0.11 to -0.02)	-0.28 (-0.47 to -0.10)

Flooding near settlements

Table A35: Further checks of the odds ratio of flooding within 500 m of water-related names (exploratory, amendment 14, items B11, B13, B14 and C1): the estimator of Eq. (2) of the paper on the capped EMS table (AOI fixed effects, distance bands, logit of the baseline model's held-out prediction, 95 % intervals clustered by location group), the planned GFM estimate with a jackknife interval, the uncapped estimate's bootstrap distribution and the check of the positional join.

Check	Odds ratio (95 %)	Cells	Flooded near
Capped EMS table (Sect. 3.4 of the paper)	1.01 (0.77–1.32)	1,489,381	361
Riverine cover as a fractional response	0.98 (0.80–1.21)	1,507,486	402
Typed AOIs only	1.00 (0.77–1.30)	1,002,686	214
With log(1 + population) and 10 m built-up share	0.98 (0.75–1.29)	1,489,381	361
Planned GFM estimate, leave-one-group-out jackknife	0.55 (0.36–0.84)	range without one group 0.51–0.63	
Observable ground only (frozen reading)	1.00 (0.76–1.33)	950,840	359
Observable ground only (edited lexicon)	1.16 (0.82–1.64)	950,840	261
Uncapped EMS, edited lexicon: resamples of the pairs bootstrap	point 0.84, median 0.87, skewness of log OR 1.4, 16 % above one		
Join of cells, features and land cover by row position	500 AOIs, 0 with different row counts; 20.6 million distances recomputed, largest difference 0.004 m		

Table A36: Diagnosis of the GFM labels (exploratory, amendment 13, items B9(e) and B14): flooded shares by AOI on the cells that GFM can assess, the baseline gain when every model is trained and scored on GFM labels, and the agreement of the two label sets by distance from the nearest settlement and by built-up land (ESA WorldCover, 10 m; more than two-thirds of these cells contain none, so the cells are split into those with and without it).

Quantity	Value
AOIs on both label sets	192
Median flooded share, EMS / GFM	5.3 % / 4.7 %
AOIs where GFM mapped less flooding than EMS	53 %
Correlation of the AOIs' flooded shares	0.81
Baseline gain on GFM labels (pooled)	-6.6×10^{-4}
Baseline gain on EMS labels (primary)	7.6×10^{-3}
AOIs with a negative baseline gain on GFM	28 %
EMS and GFM, within 500 m of the settlement	both 6.3 %, EMS only 5.6 %, GFM only 1.0 %; GFM mapped 53 % of the EMS flooding (30,376 cells)
EMS and GFM, 500 m to 1 km	both 7.8 %, EMS only 5.2 %, GFM only 1.5 %; GFM mapped 60 % of the EMS flooding (59,234 cells)
EMS and GFM, beyond 1 km	both 6.8 %, EMS only 4.9 %, GFM only 1.9 %; GFM mapped 58 % of the EMS flooding (145,093 cells)
EMS and GFM, cells without built-up land	both 8.2 %, EMS only 5.3 %, GFM only 1.9 %; GFM mapped 61 % of the EMS flooding (187,766 cells)
EMS and GFM, cells with built-up land	both 2.2 %, EMS only 4.1 %, GFM only 0.7 %; GFM mapped 35 % of the EMS flooding (46,937 cells)

Table A37: Positive control without flood labels (binary measures): within-AOI differences between cells (or settlements) whose nearest settlement name is in the class and other cells (or settlements) of the same AOI, with 95 % cluster-bootstrap intervals over location groups. "HAND $\leq$ 2 m" is whether any 25 m pixel of the 250 m cell lies within 2 m of the nearest drainage (the cell's minimum HAND); it holds for most cells, so the shares in each group are given. Lower half: the cell lies in the lower half of its AOI by mean elevation. Continuous measures are in Table A38. The all-cell rows for water-related names as a group and for highland names are the analysis planned in amendment 2; the other all-cell rows (the split by class) were computed with it and are not planned. [a]Not planned: cells within 177 m of the settlement point (the one to four cells around it), averaged per settlement (18,558 settlements; AOIs need at least two settlements of each kind). [b]Excluding settlements whose GeoNames coordinates fall on whole arc-minutes (rounding of up to about 0.9 km in each coordinate; 11 % of the settlement points of this analysis, against 14 % of the settlements nearest the analysed cells).

Nearest name	Scale	HAND $\leq$ 2 m in cell (pp)	Class/other (%)	Lower half (pp)	AOIs
water-related (any)	all cells	+0.2 [-1.0, +1.3]	82/82	+0.6 [-2.9, +4.1]	310
	at settlement[a]	+5.7 [+3.5, +8.1]	89/83	+5.0 [+1.5, +8.7]	151
water	all cells	-0.7 [-2.0, +0.4]	83/83	-4.0 [-8.9, +0.7]	176
	at settlement[a]	+3.2 [+0.1, +6.6]	88/84	-3.7 [-9.7, +2.7]	64
wetland	all cells	+0.4 [-3.3, +3.8]	85/84	+1.5 [-7.4, +11.3]	41
	at settlement[a]	+6.7 [-4.3, +22.1]	88/81	-1.5 [-18.9, +25.0]	12
riverine	all cells	+2.1 [+0.1, +4.4]	90/87	+12.8 [+5.1, +20.3]	95
	at settlement[a]	+4.7 [+0.5, +10.4]	92/87	+16.4 [+8.9, +24.5]	38
crossing	all cells	+2.1 [+0.6, +3.6]	81/79	+9.4 [+4.3, +14.5]	102
	at settlement[a]	+7.5 [+0.5, +14.5]	91/84	+9.3 [+0.8, +17.6]	34
lowland	all cells	-0.9 [-3.2, +1.2]	78/79	-4.9 [-10.4, +0.9]	160
	at settlement[a]	+5.5 [+0.4, +10.7]	85/79	+3.4 [-3.7, +9.8]	60
highland	all cells	-3.1 [-4.5, -1.6]	78/81	-7.7 [-11.7, -3.5]	253
	at settlement[a]	-10.1 [-13.4, -6.8]	72/82	-8.9 [-13.0, -4.9]	137
water-related (any)	at settlement, coordinates not rounded[b]	+6.1 [+3.7, +8.4]	90/83	+4.9 [+1.3, +8.7]	135
highland	at settlement, coordinates not rounded[b]	-10.4 [-13.9, -6.8]	71/82	-9.0 [-13.2, -4.9]	122

Table A38: Positive control without flood labels (continuous measures), as in Table A37: differences in log(1 + distance to the nearest mapped river), in the cell's mean height above nearest drainage and in its HAND percentile within the AOI. The all-cell rows for water-related names as a group and for highland names are the analysis planned in amendment 2; the other all-cell rows (the split by class) were computed with it and are not planned. [a]Not planned: cells within 177 m of the settlement point (the one to four cells around it), averaged per settlement (18,558 settlements; AOIs need at least two settlements of each kind). [b]Excluding settlements whose GeoNames coordinates fall on whole arc-minutes (rounding of up to about 0.9 km in each coordinate; 11 % of the settlement points of this analysis, against 14 % of the settlements nearest the analysed cells).

Nearest name	Scale	log(1 + river distance)	Mean HAND (m)	HAND percentile (pp)
water-related (any)	all cells	-0.04 [-0.09, +0.01]	+0.7 [-0.7, +2.3]	-0.04 [-1.6, +1.7]
	at settlement[a]	-0.22 [-0.33, -0.11]	-2.0 [-2.9, -1.3]	-4.6 [-6.1, -3.1]
water	all cells	+0.01 [-0.07, +0.08]	+1.1 [-0.3, +2.8]	+2.0 [-0.1, +4.4]
	at settlement[a]	-0.17 [-0.33, -0.01]	-0.6 [-1.7, +0.5]	-2.0 [-4.3, +0.4]
wetland	all cells	-0.001 [-0.14, +0.14]	+0.1 [-2.8, +3.7]	-3.5 [-8.2, +1.6]
	at settlement[a]	+0.17 [-0.28, +0.50]	-0.7 [-5.7, +3.9]	-1.7 [-13.6, +5.3]
riverine	all cells	-0.01 [-0.15, +0.16]	-0.6 [-3.1, +2.2]	-5.0 [-8.7, -1.3]
	at settlement[a]	-0.12 [-0.42, +0.14]	-2.2 [-4.1, -0.7]	-7.7 [-11.9, -4.4]
crossing	all cells	-0.18 [-0.27, -0.10]	-1.0 [-3.1, +0.8]	-1.5 [-4.2, +0.9]
	at settlement[a]	-0.54 [-0.86, -0.28]	-2.6 [-4.5, -1.0]	-5.3 [-9.9, -1.5]
lowland	all cells	-0.02 [-0.11, +0.07]	+1.7 [-1.0, +5.0]	+1.1 [-1.6, +3.9]
	at settlement[a]	-0.22 [-0.40, -0.05]	-2.3 [-3.9, -0.9]	-3.2 [-6.7, +0.1]
highland	all cells	+0.08 [+0.03, +0.13]	+3.4 [+1.5, +5.9]	+3.8 [+2.1, +5.5]
	at settlement[a]	+0.11 [+0.01, +0.23]	+5.3 [+3.8, +7.1]	+8.5 [+6.0, +10.8]
water-related (any)	at settlement, coordinates not rounded[b]	-0.23 [-0.35, -0.11]	-2.1 [-2.9, -1.5]	-5.0 [-6.7, -3.5]
highland	at settlement, coordinates not rounded[b]	+0.11 [-0.01, +0.22]	+5.3 [+3.7, +7.1]	+8.8 [+6.3, +11.3]

Table A39: Where settlements with water-related names stand (exploratory, amendment 8; no flood labels). Within-AOI difference between settlements whose name is in the class and other settlements of the same AOI in the height of the settlement's cell above the nearest point of a river draining at least 100 or 1000 km^2 (HydroRIVERS; mean elevation of the settlement's cell minus the lowest elevation of the river's cell, m), and in the logarithm of the drainage area of the nearest river; 95 % cluster-bootstrap intervals over location groups. Last row: the HAND of the settlement's site, for names that contain the name of a river, stream or lake within 5 km in the gazetteer.

Names	Above rivers ≥100 km^2	≥1000 km^2	log drainage area	Settlements
Water-related (pooled)	−5.0 (−9.0 to −1.3)	−4.8 (−9.0 to −1.0)	0.11 (−0.02 to 0.24)	11120
Floodplain reading (riverine, wetland)	−4.9 (−10.6 to −0.2)	−3.2 (−9.6 to 2.6)	0.28 (−0.09 to 0.67)	5150
Water bodies	0.9 (−4.5 to 6.6)	−0.9 (−6.8 to 5.1)	−0.21 (−0.48 to 0.05)	6156
Wetland	0.1 (−17.3 to 13.5)	5.3 (−17.4 to 28.2)	0.18 (−0.90 to 1.39)	1478
Riverine ground	−4.4 (−12.0 to 2.3)	−5.7 (−13.1 to 0.9)	0.31 (−0.07 to 0.71)	4077
Crossings	−8.7 (−13.9 to −3.9)	−7.5 (−13.2 to −3.0)	0.61 (0.18 to 1.13)	3121
Lowland	−6.0 (−17.3 to 4.7)	−4.3 (−16.8 to 7.4)	0.21 (−0.03 to 0.43)	5565
Names containing a nearby river or lake name (HAND of the site)	−2.4 (−4.4 to −0.6)	—	—	3931

Table A40: Within-AOI odds ratios of flooding for cells whose nearest settlement name is in the class, from logistic regressions with AOI fixed effects, unadjusted or with the logit of the held-out terrain-model prediction (M1) as a covariate; 95 % intervals from cluster-robust (location-group) standard errors; 479 AOIs (AOIs whose sampled cells are all dry carry no information and are omitted). [r] All cells, one indicator per class: the analysis planned in amendment 2. The distance-banded models (class × band, with band main effects) were not planned. The distance-band row gives the odds of a flood label 0.5–1 km and more than 1 km from the nearest settlement relative to within 500 m. Built-up and tree cover removed: cells with ESA CCI built-up cover above 10 % or tree cover above 50 % excluded. A cluster bootstrap over location groups (200 replicates, AOIs re-entered with their own intercepts) gives similar intervals, e.g. 0.73–1.28 for water-related names within 500 m given the baseline model.

Nearest name	All cells[r]		Unadjusted, by distance			Given the baseline model, by distance		
	unadjusted	given terrain	≤ 500 m	0.5–1 km	> 1 km	≤ 500 m	0.5–1 km	> 1 km
water-related (any)	1.00 [0.81, 1.23]	1.02 [0.83, 1.25]	1.05 [0.80, 1.36]	1.11 [0.91, 1.34]	0.95 [0.74, 1.22]	1.01 [0.77, 1.32]	1.06 [0.85, 1.33]	1.00 [0.80, 1.26]
water	0.85 [0.60, 1.21]	0.97 [0.73, 1.29]	0.89 [0.67, 1.20]	0.86 [0.67, 1.11]	0.85 [0.52, 1.38]	0.77 [0.57, 1.05]	0.80 [0.62, 1.03]	1.07 [0.74, 1.55]
wetland	0.98 [0.46, 2.09]	1.09 [0.47, 2.53]	1.19 [0.52, 2.73]	1.10 [0.45, 2.69]	0.92 [0.45, 1.84]	1.60 [0.71, 3.59]	1.32 [0.48, 3.62]	0.80 [0.38, 1.67]
riverine	1.30 [0.91, 1.86]	1.17 [0.74, 1.83]	1.46 [0.94, 2.26]	1.48 [1.03, 2.11]	1.16 [0.80, 1.69]	1.41 [0.92, 2.16]	1.38 [0.92, 2.08]	1.04 [0.64, 1.70]
crossing	1.02 [0.72, 1.44]	0.90 [0.65, 1.26]	0.75 [0.42, 1.35]	1.01 [0.64, 1.60]	1.04 [0.71, 1.53]	0.77 [0.46, 1.26]	0.89 [0.58, 1.35]	0.92 [0.63, 1.35]
lowland	0.83 [0.56, 1.23]	0.93 [0.65, 1.33]	0.80 [0.42, 1.54]	1.06 [0.61, 1.86]	0.76 [0.49, 1.18]	0.87 [0.50, 1.50]	1.16 [0.68, 1.97]	0.87 [0.59, 1.29]
highland	1.00 [0.83, 1.22]	1.08 [0.90, 1.30]	0.98 [0.74, 1.29]	1.04 [0.80, 1.35]	1.00 [0.81, 1.24]	1.21 [0.88, 1.68]	1.10 [0.89, 1.36]	1.07 [0.86, 1.33]
Distance band (all names)	—	—	1 (reference)	1.61 [1.48, 1.76]	1.97 [1.64, 2.37]	1 (reference)	1.12 [1.05, 1.20]	1.29 [1.12, 1.48]
water-related, built-up and tree cover removed	—	—	1.03 [0.79, 1.36]	1.09 [0.90, 1.34]	0.94 [0.73, 1.22]	1.03 [0.77, 1.37]	1.09 [0.86, 1.37]	1.01 [0.79, 1.29]

Table A41: Odds ratios of flooding near settlements by class and distance band on the EMS labels, given the baseline model's prediction (AOI fixed effects), with cluster-robust and pairs cluster bootstrap 95 % intervals over location groups (2,000 replicates; amendment 7). The two blocks are separate models (water-related names; the six classes).

Names, distance	Odds ratio	Cluster-robust	Pairs bootstrap
water-related, ≤500 m	1.01	0.77–1.32	0.77–1.31
water-related, 0.5–1 km	1.06	0.85–1.33	0.84–1.30
water-related, >1 km	1.00	0.80–1.26	0.79–1.25
water, ≤500 m	0.77	0.57–1.05	0.54–1.03
water, 0.5–1 km	0.80	0.62–1.03	0.60–1.02
water, >1 km	1.07	0.74–1.55	0.71–1.49
wetland, ≤500 m	1.60	0.71–3.59	0.41–3.62
wetland, 0.5–1 km	1.32	0.48–3.62	0.32–3.06
wetland, >1 km	0.80	0.38–1.67	0.35–1.61
riverine, ≤500 m	1.41	0.92–2.16	0.81–2.19
riverine, 0.5–1 km	1.38	0.92–2.08	0.84–2.04
riverine, >1 km	1.04	0.64–1.70	0.60–1.58
crossing, ≤500 m	0.77	0.46–1.26	0.42–1.22
crossing, 0.5–1 km	0.89	0.58–1.35	0.56–1.32
crossing, >1 km	0.92	0.63–1.35	0.59–1.29
lowland, ≤500 m	0.87	0.50–1.50	0.45–1.41
lowland, 0.5–1 km	1.16	0.68–1.97	0.61–1.83
lowland, >1 km	0.87	0.59–1.29	0.57–1.23
highland, ≤500 m	1.21	0.88–1.68	0.83–1.64
highland, 0.5–1 km	1.10	0.89–1.36	0.85–1.34
highland, >1 km	1.07	0.86–1.33	0.85–1.35

Table A42: The names increment within areas (M1c→M3n) by class of the nearest name, on the Brier scale (post hoc; planned in amendment 2 as descriptive quantities, the 95 % intervals and the difference between real and permuted names in its fourth addendum). All Brier-score differences are in units of 10^{-4}; positive values mean that names improved held-out predictions. Intervals come from the cluster bootstrap over location groups restricted to the cells of each class (classes overlap: a name can belong to several). Baseline ΔBS is the baseline gain (M0→M1) in the same cells, the denominator of the class verdicts in Table A18; the pooled equivalence margin is 3.78. [a]For all cells, the difference between real and permuted names is given as a ratio in Table S6.

Nearest name	Cells	Units	ΔBS	90 % interval	95 % interval	Real − permuted (95 %)	Baseline ΔBS
all cells (primary)	1,507,486	195	−0.264	[−0.826, +0.210]	[−0.956, +0.279]	—[a]	75.5
water-related (any)	89,404	146	−0.448	[−2.87, +1.98]	[−3.30, +2.39]	+0.895 [−3.16, +5.23]	125
water	35,814	118	+2.84	[−0.263, +6.55]	[−0.777, +7.51]	+6.33 [+0.124, +15.2]	265
wetland	2,857	51	−7.68	[−19.4, −0.00298]	[−22.6, +0.420]	−10.6 [−24.7, −1.26]	15.4
riverine	13,841	85	−14.2	[−29.6, +0.259]	[−33.2, +2.93]	−11.0 [−31.1, +8.00]	58.1
crossing	13,918	83	+2.08	[−2.65, +7.77]	[−3.61, +9.03]	+0.585 [−5.71, +8.62]	19.7
lowland	24,552	101	+2.01	[−2.43, +7.18]	[−3.26, +8.38]	+1.15 [−4.03, +6.98]	27.1
highland	58,021	130	−1.10	[−3.26, +1.12]	[−3.63, +1.54]	−0.761 [−2.79, +1.45]	86.2

Table A43: Within-AOI flood contrasts by distance from the cell to its nearest settlement (not planned; added while the planned analyses were written, with their full reporting planned in amendment 2, addendum 2): the share of cells labelled flooded nearest a water-related name minus the share nearest other names, in percentage points, observed and net of the held-out predictions of the baseline model with the regional composition of names (M1c), with 95 % cluster-bootstrap intervals over location groups. AOIs enter a band when both groups have at least 20 cells in it; they are averaged with weights equal to the size of the smaller group, or scaled to the uncapped number of evaluable cells of each AOI, which lets a few large AOIs dominate.

Distance to the nearest settlement	AOIs	Location groups	Weights: smaller group		Weights: uncapped AOI size	
			observed	net of terrain	observed	net of terrain
≤ 250 m	37	31	+1.1 [+0.04, +2.9]	+0.5 [−0.3, +1.8]	+0.1 [−1.0, +2.4]	−0.8 [−2.3, +1.6]
250–500 m	98	70	+0.2 [−0.6, +1.2]	−0.1 [−0.8, +0.9]	−1.1 [−2.6, +0.9]	−2.1 [−4.3, +0.6]
0.5–1 km	185	111	+0.4 [−0.4, +1.4]	+0.2 [−0.5, +1.1]	+0.8 [−1.7, +3.9]	−0.04 [−2.8, +2.9]
> 1 km	263	134	−0.2 [−1.1, +0.7]	−0.1 [−0.9, +0.6]	+2.9 [−0.4, +5.5]	+1.6 [−0.7, +3.4]

Table A44: Odds ratios of flooding within 500 m of settlements, given the held-out terrain prediction (AOI fixed effects, distance bands, logit of the baseline model's prediction; exploratory, amendment 8, all not planned in the protocol). Pairs: cluster bootstrap over location groups with the regression refitted, 2,000 replicates. Cleaned reading: without two-letter affixes and the English suffixes that also read transferred names (0.6 % of the cells change their reading). Uncapped: every evaluable cell of the analysed AOIs, with the predictions of the uncapped sensitivity run. Misclassification: probabilistic bias analysis with the rates measured against recorded etymologies by the model that compiled the lexicon. [s]AOIs that enter the model, which needs flooded and dry cells in an AOI, of the AOIs whose EMS map was not drawn from Sentinel-1 (capped sample; other tables count the AOIs of their own samples). Columns 4–6: flooded cells within 500 m of water-related names, location groups with any cell within 500 m of a water-related name (flooded or not), and the AOIs that enter each model (AOIs with flooded and dry cells). The uncapped row uses the readings of the frozen code; the corrected readings are in the table of uncapped estimates of amendment 11.

Estimate	Odds ratio	95 % interval	Flooded	Groups	AOIs
Pooled water-related class, EMS labels (pairs cluster bootstrap)					
All AOIs	1.01	0.77–1.31	361	134	479
AOIs with a recorded flood type	1.00	0.77–1.30	214	91	331
Without AOIs lacking a hydrography layer	0.85	0.62–1.19	183	84	269
Without Oceania and the Americas	0.96	0.71–1.29	303	107	402
Without Australia	1.00	0.74–1.30	322	124	436
Cleaned reading	1.04	0.76–1.38	271	140	479
Every evaluable cell (no cap; frozen-code readings)	0.91	0.76–1.15	3,749	137	500
Without AOIs in Pakistan	1.09	0.77–1.45	271	127	467
Without AOIs in Pakistan and Bangladesh	0.97	0.72–1.24	184	125	465
Truly water-related names (misclassification corrected)	1.00	0.63–1.63			
By the river nearest the settlement, and adjusted for its height (cluster-robust)					
Draining less than 100 km^2	0.91	0.58–1.42	188	117	431
Draining 100–1000 km^2	1.61	0.98–2.64	87	93	351
Draining at least 1000 km^2	0.84	0.52–1.36	86	89	368
Cells with a height above the main river, unadjusted	1.00	0.75–1.33	331	118	427
adjusted for the height above the river	1.00	0.75–1.33	331	118	427
Names containing a nearby river or lake name	0.88	0.58–1.33	66	112	479
By class, all classes in one model (Bonferroni 95 % interval; Holm-adjusted p in the last column)					
Water bodies	0.77	0.51–1.17			0.58
Wetland	1.60	0.54–4.76			0.94
Riverine ground	1.41	0.80–2.50			0.58
Crossings	0.77	0.39–1.50			0.94
Lowland	0.87	0.42–1.81			0.94
Highland	1.21	0.79–1.87			0.94
Riverine flood cover as a fractional outcome, water-related class by distance band					
Within 500 m	0.98	0.80–1.21			
500 m–1 km	1.05	0.86–1.27			
Beyond 1 km	1.01	0.83–1.23			
The two label sets on the same cells (pairs bootstrap; jackknife with t quantiles)					
EMS	0.78	0.60–1.07	80 groups, 176 AOIs; jackknife 0.54–1.11		
GFM	0.55	0.34–0.81	80 groups, 176 AOIs; jackknife 0.36–0.85		
Ratio of the odds ratios, GFM over EMS	0.71	0.47–0.92			
Floodplain reading, EMS	0.83	0.38–1.81	14	31	182
Floodplain reading, GFM	0.31	0.09–1.00	5	31	182
EMS map not from Sentinel-1: GFM	0.48	0.24–0.98			63 of 68[s]
EMS map not from Sentinel-1: EMS	0.77	0.46–1.28			66 of 68[s]

Table A45: Where the flooded cells within 500 m of settlements with water-related names lie (EMS labels; descriptive): 361 flooded cells among 10,048 such cells, in the AOIs of the odds ratios near settlements (AOIs with both flooded and dry cells). Share: of all flooded cells near water-related names. For the AOIs in Pakistan the fourth column gives the flooded cells of all evaluable cells (amendment 8, addendum 6). AOIs: the AOIs with such cells whose nearest settlement lies in the country or activation; an AOI whose cells are nearest settlements in several countries counts in each, so the column sums to more than the number of AOIs (510 by country).

Country or activation	Flooded	Cells	Share	AOIs
By country of the nearest settlement				
PK	90	683	25 %	12
BD	87	453	24 %	2
AU	39	728	11 %	43
DE	37	1,888	10 %	29
PT	23	315	6.4 %	9
GB	15	1,807	4.2 %	18
MX	11	224	3.0 %	3
SI	8	180	2.2 %	6
VN	6	278	1.7 %	16
ES	6	142	1.7 %	25
MM	5	192	1.4 %	11
IR	4	152	1.1 %	3
RO	3	227	0.8 %	12
HR	3	101	0.8 %	27
KH	3	37	0.8 %	4
Other 30 countries	21	2,641	5.8 %	290
The activations with the most such cells				
EMSR097: Floods in Bangladesh (2014)	87	453	24 %	2
EMSR629: Floods in Pakistan (2022)	73	149	20 %	3
EMSR047: Flood in Sachsen-Anhalt- Germany (2013)	23	199	6.4 %	2
EMSR567: Floods in Queensland- Australia (2022)	22	159	6.1 %	8
EMSR864: Flood in Portugal and Spain (2026)	22	124	6.1 %	5
EMSR838: Flood in Pakistan (2025)	17	525	4.7 %	7
EMSR570: Flood in New South Wales- Australia (2022)	14	47	3.9 %	2
EMSR845: Flood in Mexico (2025)	9	88	2.5 %	1
EMSR150: Flood in England (2015)	6	470	1.7 %	4
EMSR388: Flood in the Southeast of Spain (2019)	6	62	1.7 %	1
AOIs in Pakistan, by activation (evaluable cells; flooded near / near)				
EMSR838: Flood in Pakistan (2025)	17	525	6,047 of 38,140	7
EMSR629: Floods in Pakistan (2022)	73	149	12,068 of 20,000	3
EMSR631: Flood in Pakistan (2022)	0	9	1,165 of 3,122	2

Table A46: The pooled odds ratio of flooding within 500 m of water-related names (EMS labels, the main model of amendment 8) with each activation that has flooded cells near water-related names left out in turn (exploratory; amendment 9; 48 activations). Flooded: flooded cells near water-related names in the activation left out. Intervals: 95 %, cluster-robust over location groups.

Left out	Flooded	OR	95 %	Left out	Flooded	OR	95 %
EMSR044	2	1.01	0.77–1.33	EMSR427	2	1.00	0.77–1.31
EMSR047	23	1.00	0.76–1.32	EMSR429	2	1.01	0.77–1.32
EMSR052	2	1.02	0.78–1.33	EMSR514	3	1.01	0.77–1.32
EMSR055	1	1.01	0.77–1.32	EMSR517	2	1.01	0.77–1.32
EMSR069	4	1.03	0.78–1.35	EMSR567	22	0.98	0.74–1.28
EMSR070	4	1.01	0.77–1.32	EMSR570	14	1.01	0.77–1.34
EMSR073	1	1.01	0.77–1.32	EMSR574	3	1.01	0.77–1.32
EMSR097	87	0.91	0.72–1.14	EMSR629	73	1.08	0.81–1.44
EMSR107	4	1.00	0.76–1.31	EMSR664	1	1.01	0.77–1.32
EMSR130	5	1.04	0.80–1.35	EMSR708	1	1.01	0.77–1.32
EMSR142	1	1.01	0.77–1.32	EMSR712	4	1.01	0.77–1.32
EMSR147	3	1.01	0.77–1.33	EMSR722	5	1.01	0.77–1.32
EMSR150	6	1.00	0.76–1.31	EMSR754	3	1.02	0.78–1.34
EMSR167	1	1.01	0.77–1.32	EMSR761	1	1.01	0.77–1.32
EMSR199	1	1.00	0.77–1.32	EMSR774	2	1.01	0.77–1.32
EMSR203	2	1.01	0.77–1.32	EMSR799	2	1.01	0.77–1.32
EMSR258	1	1.01	0.77–1.32	EMSR802	2	1.01	0.77–1.32
EMSR275	1	1.01	0.77–1.32	EMSR838	17	1.01	0.76–1.34
EMSR293	1	1.01	0.77–1.32	EMSR845	9	0.98	0.75–1.29
EMSR352	1	1.02	0.78–1.33	EMSR848	3	1.02	0.78–1.34
EMSR388	6	0.99	0.76–1.30	EMSR857	1	1.01	0.77–1.32
EMSR407	2	1.00	0.77–1.31	EMSR861	1	1.01	0.77–1.32
EMSR419	3	1.00	0.77–1.32	EMSR864	22	0.97	0.74–1.28
EMSR424	3	1.01	0.77–1.33	EMSR866	1	1.01	0.77–1.32

Table A47: The six class coefficients of the model with all lexicon classes within 500 m (EMS labels; the figure of odds ratios near settlements in the paper): unadjusted p, Holm-adjusted p (Holm, 1979) and Benjamini–Hochberg q (Benjamini and Hochberg, 1995) (exploratory; amendment 9).

Class	p	Holm p	BH q
Water bodies	0.097	0.584	0.339
Wetland	0.258	0.942	0.356
Riverine ground	0.113	0.584	0.339
Crossings	0.297	0.942	0.356
Lowland	0.609	0.942	0.609
Highland	0.236	0.942	0.356

Table A48: Odds ratio near water-related names corrected for misreading over a grid of rates (exploratory, amendment 10, item B5): $\mathrm{OR} = (\mathrm{OR}^*(1-p_0)-(1-p_1))/(p_1-\mathrm{OR}^*p_0)$, with OR^* the observed estimate within 500 m and the limits of its pairs-bootstrap 95 % interval; p_1: share of water-related names among those read as water-related; p_0: share among the others. Undefined: the observed value cannot arise from these rates.

p_1	p_0	Observed point	Observed lower	Observed upper
0.6	0.05	1.02	0.60	1.57
0.6	0.10	1.02	0.57	1.64
0.6	0.20	1.02	0.49	1.89
0.6	0.30	1.03	0.39	2.45
0.7	0.05	1.01	0.66	1.48
0.7	0.10	1.02	0.64	1.53
0.7	0.20	1.02	0.59	1.69
0.7	0.30	1.02	0.52	1.98
0.8	0.05	1.01	0.70	1.41
0.8	0.10	1.01	0.69	1.45
0.8	0.20	1.02	0.65	1.56
0.8	0.30	1.02	0.60	1.74
0.9	0.05	1.01	0.74	1.36
0.9	0.10	1.01	0.73	1.39
0.9	0.20	1.01	0.70	1.47
0.9	0.30	1.02	0.66	1.59

Table A49: Odds ratio of flooding within 500 m of settlements with water-related names against other names of the same AOI, given the held-out prediction of the baseline model (exploratory; amendment 11, items B2 and B5): AOI fixed effects, distance bands and the logit of the prediction, with a 95 % pairs cluster bootstrap over location groups (2,000 replicates). Flooded: flooded cells within 500 m of water-related names and the location groups that hold them. MDE: the smallest odds ratios below and above one detectable with 80 % power (two-sided 5 %). The uncapped sample conditions on the predictions of the uncapped sensitivity run (the baseline model does not use names). The GFM labels are not independent of EMS where the EMS map was drawn from Sentinel-1. The number of AOIs not mapped from Sentinel-1 differs between tables and figures because each counts the AOIs that enter its own sample: uncapped or capped, cells that GFM can assess, and AOIs with both flooded and dry cells.

Sample and labels	Odds ratio	95 % interval	Flooded	Groups	MDE
Every evaluable cell (no cap), corrected-code readings					
EMS labels, 500 AOIs	0.91	0.76–1.16	3,753	75	0.74, 1.35
GFM labels, cells that GFM can assess	0.82	0.60–1.07	434	29	0.66, 1.50
EMS labels on the same cells	0.89	0.67–1.16	560	36	0.68, 1.47
GFM labels, AOIs not mapped from Sentinel-1 (111)	0.85	0.43–1.46	166	15	0.44, 2.29
EMS labels, the same AOIs and cells	1.00	0.71–1.58	263	20	0.56, 1.79
EMS labels, flooded fraction over observed non-permanent water	0.91	0.77–1.17	3,810	76	0.74, 1.35
Capped table (10,000 cells per location group)					
EMS labels	1.01	0.77–1.31	361	52	—
EMS labels, flooded fraction over observed non-permanent water	1.02	0.78–1.32	372	54	0.68, 1.46

Table A50: The floodplain reading (riverine and wetland classes) within 500 m of settlements, given the terrain prediction (exploratory, amendment 10, item B2; defined after the class estimates had been seen). Intervals: pairs cluster bootstrap over location groups, 2,000 replicates. Countries are those of the nearest settlement. ^{x}Forms removed in the re-reading. Columns 4–5: flooded cells and location groups with such cells (or cells, for the forms).

Estimate	Odds ratio	95 % interval	Flooded	Groups
All AOIs (floodplain reading)	1.45	0.93–2.08	163	85
re-read through the same code, all forms	1.42	0.90–2.02	163	91
re-read without bank, au, dike and polder forms	1.54	0.94–2.28	142	85
Without activation EMSR097	1.39	0.75–2.28	82	83
One country left out				
without BD (81 flooded cells)	1.39	0.75–2.23	82	83
without DE (22 flooded cells)	1.55	0.92–2.31	141	79
without AU (16 flooded cells)	1.35	0.83–1.84	147	77
without MX (9 flooded cells)	1.38	0.86–2.00	154	83
without PK (8 flooded cells)	1.55	0.98–2.25	155	80
without SI (8 flooded cells)	1.40	0.87–1.99	155	83
without ES (6 flooded cells)	1.40	0.86–2.03	157	80
without GB (3 flooded cells)	1.46	0.90–2.10	160	77
without RO (3 flooded cells)	1.44	0.88–2.10	160	82
without BE (2 flooded cells)	1.44	0.91–2.07	161	85
without KH (2 flooded cells)	1.45	0.92–2.08	161	85
without AL (1 flooded cell)	1.44	0.90–2.05	162	84
without MM (1 flooded cell)	1.45	0.93–2.05	162	84
without VN (1 flooded cell)	1.48	0.93–2.12	162	79
Composition by matched form (flooded cells within 500 m; top 12)				
riverine *char*			71	340 cells
riverine *au*^{x}			19	459 cells
riverine *island*			14	27 cells
wet *bil*			9	59 cells
wet *estero*			9	18 cells
riverine *otok*			8	34 cells
riverine *arenal*			6	19 cells
wet *kallar*			5	7 cells
riverine *lunca*			3	74 cells
riverine *bela*			3	23 cells
riverine *kaoh*			2	10 cells
riverine *flat*			2	49 cells

Table A51: Calibration of the held-out predictions (exploratory, amendment 8): intercept and slope of a logistic regression of the labels on the logit of the prediction, Brier score with the reliability and resolution terms of the decomposition of Murphy (1973) (ten equal-count bins), and median within-AOI area under the ROC curve. M1: baseline model; M1s: recalibrated; M1c: with the regional composition of names (reference of the primary comparison); M3n: with the names. Skill: each AOI predicted flooded on as many cells as the labels flood (CSI: critical success index). The primary, country, time and language-model splits use the confirmatory run (frozen code); the Brier scores of the corrected run, which differ in the fourth decimal, are in Table A23.

Split	Model	Intercept	Slope	Brier	Reliability	Resolution	AUC
Primary	M1	-0.27	0.85	0.0437	3.4×10^{-4}	7.7×10^{-3}	0.90
	M1s	0.09	1.05	0.0434	5.9×10^{-5}	7.7×10^{-3}	0.90
	M1c	-0.23	0.93	0.0431	2.1×10^{-4}	8.0×10^{-3}	0.90
	M3n	-0.23	0.93	0.0431	2.2×10^{-4}	8.0×10^{-3}	0.90
Countries held out	M1	-0.57	0.69	0.0494	1.6×10^{-3}	5.5×10^{-3}	0.90
	M1s	-0.18	0.95	0.0479	6.7×10^{-4}	5.4×10^{-3}	0.90
	M1c	-0.39	0.84	0.0485	8.1×10^{-4}	5.5×10^{-3}	0.90
	M3n	-0.39	0.84	0.0485	7.9×10^{-4}	5.5×10^{-3}	0.90
Floods from 2024	M1	-0.88	0.75	0.0472	2.9×10^{-3}	3.4×10^{-3}	0.90
	M1s	-0.58	0.95	0.0460	2.0×10^{-3}	3.4×10^{-3}	0.90
	M1c	-0.70	0.91	0.0468	2.6×10^{-3}	3.5×10^{-3}	0.90
	M3n	-0.71	0.91	0.0468	2.6×10^{-3}	3.5×10^{-3}	0.90
After the LM cut-off	M1	-0.69	0.80	0.0499	1.9×10^{-3}	4.2×10^{-3}	0.90
	M1s	-0.44	0.97	0.0494	1.5×10^{-3}	4.2×10^{-3}	0.90
	M1c	-0.53	0.97	0.0497	2.0×10^{-3}	4.3×10^{-3}	0.90
	M3n	-0.54	0.97	0.0498	2.0×10^{-3}	4.3×10^{-3}	0.91
Events held out whole, frozen code	M1	-0.56	0.70	0.0488	1.4×10^{-3}	5.3×10^{-3}	0.90
	M1s	-0.13	0.96	0.0477	3.7×10^{-4}	5.1×10^{-3}	0.90
	M1c	-0.69	0.74	0.0518	2.1×10^{-3}	5.4×10^{-3}	0.90
	M3n	-0.68	0.74	0.0516	2.1×10^{-3}	5.4×10^{-3}	0.90
Events held out whole, corrected code	M1	-0.56	0.70	0.0488	1.4×10^{-3}	5.3×10^{-3}	0.90
	M1s	-0.13	0.96	0.0477	3.7×10^{-4}	5.1×10^{-3}	0.90
	M1c	-0.70	0.74	0.0519	2.2×10^{-3}	5.4×10^{-3}	0.90
	M3n	-0.70	0.74	0.0518	2.2×10^{-3}	5.4×10^{-3}	0.90
GFM labels, refitted	M1	-1.13	0.38	0.0858	6.6×10^{-3}	3.5×10^{-3}	0.86
	M1s	-0.48	0.79	0.0778	8.2×10^{-4}	2.9×10^{-3}	0.86
	M1c	-1.99	0.16	0.1155	2.5×10^{-2}	2.2×10^{-3}	0.86
	M3n	-1.99	0.16	0.1150	2.5×10^{-2}	2.2×10^{-3}	0.85

Skill in flood-mapping terms, primary split, median over AOIs (interquartile range)

Baseline model (M1)	CSI 0.13 (0.03–0.27); hit rate 0.23 (0.07–0.42); false alarms 0.77 (0.58–0.93); AUC 0.90 (0.80–0.96); 479 AOIs
HAND alone (mean HAND of the cell; corrected run)	CSI 0.08 (0.00–0.21); hit rate 0.16; false alarms 0.84; 479 AOIs

How the names were read

Table A52: Checks of the lexicon and of the readings (exploratory, amendment 13, items B10 to B12 and C2). Counts are of the settlements nearest to at least one evaluable cell of the uncapped table, read with the frozen lexicon. Forms are matched in the ASCII field of GeoNames; Unicode names are read with umlauts, å, ø folded to one letter, æ to ae and ß to ss. Both coders' codes are in `docs/verification/v70_two_coders.csv` of the code repository.

Quantity	Value
Analysed settlements matched by homographs and wrong-language readings	
German *-ach* not in *-bach*	57
German *-ach* in *-bach*	427
Norwegian *-ore*, *-ora*	2
of them in *Store-*, *Stora-*	0
French *mont-* as a prefix	228
of them *Montreuil*, *Montier*, *Moutier*	12
German *Ried*	75
Punjabi/Urdu *kacha*	1
Burmese *taung*	139
Belgian *-eau* read as German *-au*	9
Italian names matched by the German block	5
Iran, *ab* as a word	45
Iran, *-ab* as an ending	48
Finnish *Ala*	1
Finnish *Yli*, *Ylä*	1
Edited lexicon (amendment 13, item C2): entries removed or added (form × position)	
Two-letter affixes and English transfer endings	15
Springs and wells	145
Rapids and waterfalls	10
Homographs	15
Relative qualifiers	181
Riverine entries named as ordinary meadows, islands or sand	37
Added (named omissions)	13
Entries before / after	2,538 / 2,148
Water-related readings gained / lost when the Unicode name is read	
bg	0 / 1
de	4 / 0
fi	4 / 0
km	0 / 57
pl	0 / 2
sv	17 / 0
tr	0 / 3
Agreement of the two coders on water-related derivations (κ)	0.91
Codes deposited (names in an audit sample left out)	360 of 462
Language-model reading by lexicon block: settlements, water-related, unknown	
fr	13,745, 15 %, 2.0 %
km	10,702, 39 %, 1.1 %
my	9,547, 69 %, 1.8 %
de	9,463, 21 %, 0.6 %
en	8,963, 20 %, 2.4 %
it+de	6,552, 16 %, 1.3 %
es+nah	5,569, 13 %, 0.8 %
fa+hi	5,332, 9.9 %, 4.1 %

Table A53: The multilingual lexicon by block: languages covered, countries mapped to the block, number of distinct forms per class (water, wetland, riverine, crossing, lowland, highland), match positions used and number of spring and well forms removed in the strict reading. Countries are mapped to the blocks of their main languages; regional languages are not read.

Block	Languages	Countries	Wa	We	Ri	Cr	Lo	Hi	Positions	Spr
en	English	AU BW CA CM ET FJ GB GH GY IE IM KE LK LR LS MT MW NA NG NZ PG PH SG SL SS SZ TZ UG US ZA ZM ZW	25	16	16	9	18	25	word prefix suffix	6
de	German	AT BE CH DE IT LI LU	11	12	11	12	8	15	word prefix suffix	5
nl	Dutch	BE NL SR	9	7	11	5	4	6	word prefix suffix	0
fr	French	BE BF BI BJ CA CD CF CG CH CI CM DJ DZ FR GN HT LU MA MC MG ML NE RW SN TD TG TN	15	16	15	8	12	18	word prefix suffix	5
it	Italian	CH IT SM VA	19	16	14	6	13	16	word prefix	6
es	Spanish	AD AR BO CL CO CR CU DO EC ES GT HN MX NI PA PE PR PY SV UY VE	16	22	16	6	13	22	word	6
pt	Portuguese	AO BR CV GW MZ PT TL	18	17	12	7	8	14	word	5

Block	Languages	Countries	Wa	We	Ri	Cr	Lo	Hi	Positions	Spr
el	Greek (romanized)	CY GR	24	8	3	5	5	12	word prefix	11
sh	Serbo-Croatian / Bosnian (Latin)	BA HR ME RS XK	14	11	7	6	7	14	word	7
sl	Slovene	SI	7	6	5	2	5	12	word	3
bg	Bulgarian / Macedonian (romanized)	BG MK	9	6	3	2	7	13	word	6
sq	Albanian	AL MK XK	13	8	4	5	7	11	word	4
ro	Romanian	MD RO	17	10	10	8	9	14	word	5
hu	Hungarian	HU	8	6	4	4	4	7	word prefix suffix	2
cs	Czech / Slovak	CZ SK	15	13	8	6	8	15	word	5
pl	Polish	PL	13	15	12	5	6	14	word	4
uk	East Slavic (Ukrainian / Russian / Belarusian, romanized)	BY GE KG KZ MD RU TJ TM UA UZ	15	12	15	8	14	19	word	6
lt	Lithuanian	LT	9	6	6	4	5	11	word suffix	2
lv	Latvian	LV	7	4	6	3	4	5	word suffix	2
et	Estonian	EE	7	5	4	3	3	5	word suffix	1
fi	Finnish	FI	10	8	4	3	4	10	word suffix	1
sv	Swedish	FI SE	10	7	9	6	6	16	word suffix	2
no	Norwegian (also used for Icelandic placeholders)	IS NO	16	8	14	6	6	20	word suffix	1
da	Danish	DK	6	4	5	5	4	8	word suffix	1
tr	Turkish	CY TR	11	5	3	3	6	10	word prefix suffix	4
ar	Arabic (romanized; Levant, Gulf, Maghreb conventions)	AE DJ DZ EG ER IL IQ JO LB LY MA OM PS SA SD SO SY TD TN YE	19	12	8	6	13	16	word	8
fa	Persian / Urdu / Pashto / Sindhi (romanized)	AF IR PK TJ	26	8	11	7	14	16	word prefix suffix	4
hi	Hindi / Bengali / Nepali (romanized)	BD IN NP PK	17	8	11	4	8	15	word suffix	0
vi	Vietnamese (asciiname; diacritics stripped — only low-collision syllables)	VN	7	3	2	3	1	2	word	1
id	Indonesian / Malay	ID MY SG TL	13	9	7	7	6	8	word prefix	1
tl	Filipino / Tagalog and related	PH	7	4	5	4	4	7	word	1
sw	Swahili	BI CD KE RW TZ UG	7	5	2	4	2	6	word	2
bantu	Southern/Central Bantu (Chichewa/Nyanja, Shona, Sena, Zulu/Xhosa, Sotho; broad set)	AO BW CD LS MW MZ SZ ZA ZM ZW	15	4	2	4	3	10	word prefix	1
af	Afrikaans	NA ZA ZW	6	5	3	3	5	7	word suffix	2
mg	Malagasy	MG	6	4	3	2	4	10	word prefix	1
qu	Quechua / Aymara (Andes)	BO EC PE	11	2	0	0	2	9	word suffix	2
nah	Nahuatl (Mexico)	MX	3	0	0	0	0	3	word suffix	1
my	Burmese (romanized)	MM	4	0	1	0	0	3	word	0
th	Thai / Lao (romanized)	LA TH	7	4	2	2	3	7	word	0
km	Khmer (romanized)	KH	9	0	4	3	0	4	word	0
zh	Chinese (pinyin; only low-ambiguity syllables)	CN TW	2	0	0	1	0	1	word suffix	0
ka	Georgian (romanized)	GE	5	1	0	1	3	4	word suffix	2
kk	Kazakh / Kyrgyz / Uzbek / Turkmen / Mongolian (romanized; Turkic core)	KG KZ MN TM UZ	12	5	5	5	5	9	word suffix	2
so	Somali	DJ SO	7	3	1	2	3	4	word	1
am	Amharic / Tigrinya (romanized)	ER ET	7	2	0	2	4	5	word	2
ng	Hausa / Yoruba / Igbo (Nigeria, Niger)	NE NG	8	3	0	2	1	6	word prefix	0
tupi	Tupi-derived Brazilian toponymy (prefixes only, high-precision)	BR	7	0	0	0	0	3	word prefix	0

Table A54: The 20 water-related forms that matched the nearest name of the most analysed cells (amendment 6): language block, form, position, class, number of cells and of settlements. Overlapping forms are each counted: every name in *-bach* also matches *-ach*.

Block	Form	Position	Class	Cells	Settlements
en	*creek*	word	water	4,979	62
fa	*chah*	word	water	4,665	110
en	*bridge*	word	crossing	4,605	142
en	*lake*	word	water	3,002	35
de	*ach*	suffix	water	2,891	404
de	*au*	suffix	riverine	2,701	182
de	*bach*	suffix	water	2,541	360
en	*dale*	suffix	lowland	2,515	56
hi	*char*	word	riverine	2,303	149
en	*ford*	suffix	crossing	2,161	74
en	*lower*	word	lowland	1,679	39
de	*nieder*	prefix	lowland	1,260	99
en	*vale*	word	lowland	1,234	15
ro	*lunca*	word	riverine	1,211	7
en	*flat*	word	riverine	1,092	7
sv	*dalen*	suffix	lowland	1,067	11
en	*burn*	suffix	water	1,038	30
en	*well*	suffix	water	996	47
pt	*vale*	word	lowland	881	95
no	*foss*	suffix	water	820	8

Table A55: Audit of matched forms (exploratory, amendment 8): forms that the lexicon's own rules excluded (two-letter affixes), English suffixes that also read transferred and personal names, and regional words, with the number of analysed settlements (94,524 in all) whose name they match, the cells nearest those settlements, the countries and examples (most cells first).

Block	Form	Settlements	Cells	Countries	Examples
de	*au* (suffix)	182	2,701	DE 162, AT 12, BE 8	Choerau, Sandau, Obslau, Trebbichau
sv	*as* (suffix)	158	3,282	SE 156, FI 2	Fryksas, Neistenkangas, Svartnas, Marnas
no	*as* (suffix)	3	127	NO 3	Kjonnas, Raipas, Gevingas
fa	*ab* (suffix)	73	589	PK 40, IR 33	Arab, Goth Qutab Brahui, Haji Tayyab, Haji Gulab
en	*burn* (suffix)	30	1,038	GB 22, US 6, AU 2	Heyburn, Woodburn, Wythburn, Blackburn
en	*bourne* (suffix)	5	99	GB 4, IE 1	Cranbourne, Pangbourne, Barbourne, Nailsbourne
en	*borne* (suffix)	1	30	AU 1	Tichborne
en	*pool* (suffix)	5	107	GB 4, US 1	Glenpool, Sampool, Bathpool, Firepool
en	*well* (suffix)	47	996	GB 26, US 14, IE 6, AU 1	Brauncewell, Howell, Greenwell Point, Cranwell
en	*wells* (suffix)	1	138	GB 1	Kingswells
hi	*char* (word)	149	2,303	BD 149	Char Kodalia, Char Gobindapur, Char Gazi Para, Sonatani Char
fa	*bela* (word)	9	200	PK 9	Bela Ali Khanana, Kot Bela, Bela Bhagwanpura, Makki Bela
fa	*kacha* (word)	1	11	PK 1	Kacha Pakka Mor
fa	*pattan* (word)	3	133	PK 3	Pattan Khurd, Bahkan Pattan, Sahuka Pattan
fa	*pattan* (prefix)	1	10	PK 1	Chah Pattanwala
fa	*tibba* (word)	9	79	PK 9	Tibba, Tibba Ghhale, Tibba Hussain Shah, Gharianwala Tibba

Most frequent matched forms by country (class; settlements / cells)

AU	*creek* (water; 33/4,374); *lake* (water; 9/2,573); *hill* (high; 30/2,442); *bridge* (crossing; 8/1,468); *dale* (low; 14/1,377)
DE	*ach* (water; 392/2,809); *berg* (high; 417/2,715); *au* (riverine; 162/2,642); *bach* (water; 354/2,530); *ober* (high; 213/1,419)
GB	*hill* (high; 142/3,049); *bridge* (crossing; 47/1,900); *ford* (crossing; 50/1,219); *dale* (low; 28/997); *ridge* (high; 25/860)
SE	*as* (high; 158/3,186); *berg* (high; 70/2,198); *asen* (high; 19/1,876); *berget* (high; 16/1,409); *dalen* (low; 14/1,080)
US	*hill* (high; 45/694); *creek* (water; 28/604); *valley* (low; 22/565); *ford* (crossing; 29/519); *ridge* (high; 31/443)
IT	*monte* (high; 65/1,277); *ponte* (crossing; 80/755); *piana* (low; 3/629); *poggio* (high; 21/561); *sotto* (low; 19/423)
FR	*mont* (high; 206/1,710); *haut* (high; 30/379); *pont* (crossing; 61/331); *haute* (high; 14/295); *motte* (high; 12/278)
PT	*monte* (high; 46/1,648); *vale* (low; 93/849); *cima* (high; 57/552); *agua* (water; 10/388); *porto* (crossing; 31/356)
NO	*foss* (water; 8/820); *berget* (high; 4/804); *berg* (high; 10/698); *dal* (low; 6/542); *botn* (low; 2/335)
RO	*lunca* (riverine; 7/1,211); *sus* (high; 9/818); *jos* (low; 10/744); *deal* (high; 7/715); *valea* (low; 9/503)
IR	*chah* (water; 35/3,421); *bala* (high; 15/396); *nahr* (water; 17/243); *koh* (high; 10/234); *ab* (water; 62/209)
BO	*loma* (high; 5/821); *pampa* (low; 11/531); *puentes* (crossing; 1/528); *pata* (high; 8/414); *baja* (low; 14/397)
ES	*arriba* (high; 2/285); *rio* (water; 12/284); *arroyo* (water; 3/255); *sierra* (high; 5/246); *cuesta* (high; 4/205)
MG	*ambohi* (high; 40/1,879); *ambato* (high; 14/628); *andrano* (water; 20/407); *vohi* (high; 4/261); *berge* (riverine; 1/7)
PE	*cerro* (high; 2/934); *alto* (high; 7/852); *cocha* (water; 3/336); *pampa* (low; 2/241); *loma* (high; 4/229)

Table A56: Examples of names read as water-related by the lexicon, with the derivation recorded on Wikipedia (amendment 6): for each class, the two lowest-numbered coded names whose recorded derivation matches the class and the lowest-numbered one whose derivation does not. Quotations abridged; sources in the code repository.

Class	Agreement	Name	Recorded derivation	Quotation
water	agrees	Windermere (GB)	water, other	"Windermere" is thought to translate as "Winand or Vinand's lake" ... 'mere', meaning 'lake' or 'pool'
water	agrees	Odolena Voda (CZ)	other, water	The name means "Odolen's water" ... named after Knight Odolen ... refers to the sufficient water in the area
water	disagrees	New Wells (US)	other	Austrian settlers referred to the community as Neu Wels, after the city in Austria, which was later Americanized to New Wells
wetland	agrees	Spoerkelnbruch (DE)	other, wet	Vermutlicher Namensgeber ist der Faulbaum ... Bruch ist eine alte Bezeichnung für ein Moor oder Feuchtgebiet
wetland	agrees	Clairmarais (FR)	wet	l'oïl clair + maresc : le « marais aux eaux claires »
wetland	disagrees	Spodnje Blato (SI)	low	Spodnje Blato (literally 'lower Blato')
riverine	agrees	L'Ile-Saint-Denis (FR)	riverine, other	L'Île-Saint-Denis doit son nom à l'abbaye de Saint-Denis ... qui eut seigneurie sur l'île centrale
riverine	agrees	Ostrvo (RS)	riverine	"Ostrovo" means "island" in Serbian. This name originates from the fact that Ostrovo is located on former Danube island
riverine	disagrees	Veliki Otok (SI)	water	Voda (reka Pivka) pa naj bi odtekala v Postojnsko jamo ... odtok se imenuje Veliki odtok ... dali vasi ime Veliki Otok
crossing	agrees	Pont-Pean (FR)	crossing, other	le toponyme Pont-Péan peut donc signifier « pont paysan » ou « pont construit ou possédé par un dénommé Péan »
crossing	agrees	Saint-Fargeau-Ponthierry (FR)	crossing, other	Saint-Fargeau est une déformation du nom de saint Ferréol ... Ponthierry trouverait son origine dans un pont sur l'École franchi par Thierry
crossing	disagrees	Ennerdale Bridge (GB)	low, other, water	The name Ennerdale means 'Anund's valley' ... cross- influenced by 'Ehen', the name of the local river
lowland	agrees	Dol pri Ljubljani (SI)	low	The name is derived from the Slovene common noun dol 'small valley'
lowland	agrees	Spodnji Velovlek (SI)	low	The name Spodnji Velovlek literally means 'lower Velovlek'
lowland	disagrees	Germigny-sous-Coulombs (FR)	other	Du latin germinare, « germer », du latin germinius, « rejet (botanique) » ou « rejeton, progéniture ». Coulombs fait référence à Coulombs-en-Valois

Table A57: The readings in more detail (amendment 6). Precision against derivations recorded on Wikipedia by the position of the matched water-related form and by class (a name can match at more than one position, so the position rows overlap and their counts sum to more than the names); recall by class is unweighted and therefore not comparable with the weighted recall reported in the paper. Lower part: readings by continent and languages that are not read.

Quantity	Count	Share
Precision, matched as a whole word	51/71	72 %
Precision, matched as an ending	25/30	83 %
Precision, matched as a beginning	7/8	88 %
Class water: precision / recall (unweighted)	31/38 / 31/57	82 % / 54 %
Class wetland: precision / recall (unweighted)	3/6 / 3/15	50 % / 20 %
Class riverine: precision / recall (unweighted)	8/11 / 8/13	73 % / 62 %
Class crossing: precision / recall (unweighted)	16/18 / 16/21	89 % / 76 %
Class lowland: precision / recall (unweighted)	18/37 / 18/34	49 % / 53 %
Class highland: precision / recall (unweighted)	8/15 / 8/40	53 % / 20 %
Strict reading: precision / recall (weighted)	—	83 % / 19 %
Water codes reclassified as springs or wells only	9	—
Cells whose nearest name is water-related, Africa: language model / lexicon	144,265 cells	44 % / 0.5 %
Cells whose nearest name is water-related, Asia: language model / lexicon	196,090 cells	31 % / 6.7 %
Cells whose nearest name is water-related, Europe: language model / lexicon	824,854 cells	37 % / 6.3 %
Cells whose nearest name is water-related, North America: language model / lexicon	85,031 cells	17 % / 6.8 %
Cells whose nearest name is water-related, Oceania: language model / lexicon	150,806 cells	53 % / 11 %
Cells whose nearest name is water-related, South America: language model / lexicon	106,440 cells	30 % / 3.1 %
Cells nearest a settlement in ES-Catalonia	10,000	0.7 %
Cells nearest a settlement in ES-Basque Country	174	0.0 %
Cells nearest a settlement in ES-Galicia	1,528	0.1 %
Cells nearest a settlement in ES-Valencia	20,346	1.3 %
Cells nearest a settlement in ES-Navarre	1,113	0.1 %
Cells nearest a settlement in FR-Brittany	8,229	0.5 %
Cells nearest a settlement in FR-Corsica	9,905	0.7 %
Cells nearest a settlement in IE	19,576	1.3 %
Cells nearest a settlement in AU	145,844	9.7 %
Cells whose nearest name has a riverside construction	—	0.3 %

Table A58: Recorded etymologies on Wikipedia by lexicon class and by language of the name (post hoc, amendment 5). Upper part: among the names in each class of the lexicon whose Wikipedia section states a derivation, the share whose recorded derivation is in the same class and the share whose derivation is water-related in any class (the highland class is not water-related). Lower part, for languages with at least 30 coded sections: the share of water-related derivations among the names that the lexicon read as water-related (its precision) and among the other names, with counts. Shares are unweighted; intervals are 95 % (Wilson, 1927).

Lexicon class	Coded	Same class	Water-related, any class
water	38	82 % (67–91)	84 % (70–93)
wetland	6	50 % (19–81)	100 % (61–100)
riverine	11	73 % (43–90)	82 % (52–95)
crossing	18	89 % (67–97)	94 % (74–99)
lowland	37	49 % (33–64)	54 % (38–69)
highland	15	53 % (30–75)	27 % (11–52)
Language of the name	**Coded**	**Water-related, read as water-related**	**Water-related, others**
English	48	75 % (55–88); 18/24	29 % (15–49); 7/24
French	120	67 % (50–80); 22/33	18 % (12–28); 16/87
Polish	35	40 % (12–77); 2/5	3 % (1–17); 1/30
Slovene	54	94 % (73–99); 16/17	16 % (8–31); 6/37

Table A59: Classes that the language-model reading (N-read) assigned to the nearest names of the analysed cells (exploratory; amendment 10, item B15), counted once per name and weighted by the number of analysed cells nearest each name. “Unknown”: the model gave no class.

Class	Names	Share	Cells	Share
water	12,072	13 %	234,593	16 %
wetland	7,602	8.0 %	125,945	8.4 %
riverine	7,716	8.2 %	105,543	7.0 %
crossing	1,783	1.9 %	29,676	2.0 %
lowland	3,472	3.7 %	61,728	4.1 %
highland	8,832	9.3 %	161,156	11 %
other environmental	10,937	12 %	148,830	9.9 %
non-environmental	38,219	40 %	550,235	37 %
unknown	3,891	4.1 %	89,780	6.0 %

Table A60: The nearest settlements of the analysed cells by GeoNames feature code (exploratory; amendment 11, item B9): settlements, their share, those read as water-related and the cells nearest them. PPL: populated place; PPLA–PPLA4: seats of administrative divisions; PPLX: section of a populated place; PPLH: historical; PPLQ: abandoned; PPLW: destroyed; PPLL: locality; PPLF: farm village; PPLR: religious.

Feature code	Settlements	Share	Water-related	Cells
PPL	84,867	90 %	5369	1,290,881
PPLX	3,907	4.1 %	278	70,588
PPLL	1,807	1.9 %	178	37,817
PPLA3	1,669	1.8 %	59	54,253
PPLQ	633	0.7 %	33	7,390
PPLA2	598	0.6 %	26	25,149
PPLA4	579	0.6 %	84	12,093
PPLF	165	0.2 %	16	1,726
PPLA	159	0.2 %	9	5,234
PPLH	82	0.1 %	3	1,780
PPLCH	20	0.0 %	0	151
PPLW	18	0.0 %	1	138
PPLR	11	0.0 %	0	172
PPLC	4	0.0 %	0	102
PPLS	4	0.0 %	0	10
PPLG	1	0.0 %	0	2
Settlements whose coordinates are whole arc-minutes: 14 %				

Table A61: Settlements whose namesake is recorded on Wikidata (property P138), classed by the type of the namesake (amendment 5). The lexicon contains no river names by design, so it reads few settlements named after a river; most recorded namesakes are in France.

Quantity	Count
Analysed settlements	94,524
with a Wikidata item	26,509
with a recorded namesake (P138)	299
named after a water feature	139
named after a person, event or other commemoration	97
Water-feature namesakes read as water-related by the lexicon	9 of 139 (6 %)
Same, by the language model	66 of 139 (47 %)
Lexicon water-related names with a water-feature namesake	9 of 13 (69 %)
Language-model water-related names with a water-feature namesake	66 of 92 (72 %)
Lexicon water-related names with a commemorative namesake	2 of 13 (15 %)

Table A62: Precision and recall of the lexicon's water-related reading by language of the article, for languages with at least 30 coded names (amendment 5; recall by language, amendment 7, addendum 1). Recall is estimated as a domain with the design weights of the whole sample (stratum size over sample size), whereas Table A63 weights the strata by their sizes within each country, with percentile intervals from 2,000 resamples within strata. The derivations were coded by the language model that compiled the lexicon. Strict: the strict reading (water without springs and wells, wetland, riverine) against strict derivations (amendment 8, addendum 7).

Language	Coded names	Precision (95 % interval)	Strict precision	Recall (95 % interval)
fr	120	22 of 33 (67 %; 50 %–80 %)	5 of 7 (36 %–92 %)	17 % (10 %–28 %)
sl	54	16 of 17 (94 %; 73 %–99 %)	8 of 9 (56 %–98 %)	29 % (15 %–59 %)
en	48	18 of 24 (75 %; 55 %–88 %)	9 of 11 (52 %–95 %)	28 % (16 %–56 %)
pl	35	2 of 5 (40 %; 12 %–77 %)	1 of 1 (21 %–100 %)	23 % (0 %–100 %)

Table A63: Precision and recall of the lexicon's water-related reading by country, against etymologies recorded on Wikipedia (exploratory, amendment 10, item B6): countries with at least ten coded names. Recall weights the two sampling strata by their sizes within the country, whereas Table A62 uses the design weights of the whole sample (stratum size over sample size); the two agree on precision and differ on recall where the sampling fraction differs between countries. Forms: the only matched forms with at least ten coded names read as water-related. Both coders are language models; coder 1 compiled the lexicon.

Country	Coded	Read as water-related	Precision (1)	Recall (1)	Precision (2)	Recall (2)
FR	120	33	67 %	12 %	61 %	7.9 %
SI	54	17	94 %	21 %	85 %	19 %
PL	35	5	40 %	17 %	80 %	12 %
GB	33	16	81 %	20 %	83 %	23 %
CZ	20	6	83 %	33 %	83 %	20 %
MX	10	8	88 %	100 %	88 %	100 %
Form *sous* (coder 1)		12	25 %			
Form *sous* (coder 2)		11	27 %			

Table A64: Analysed cells whose names are read in a language other than their own (exploratory, amendment 10, item B14). Country level: the first language of the GeoNames country record is not among the lexicon blocks applied to the country. Regions: Natural Earth first-level units (point in polygon of the cell centre).

Country or region	Cells	Water-related	Flooded, 500 m	AOIs
Countries whose first language is not among the lexicon blocks applied				
PK (ur; read as fa, hi)	61,219	2,249	4534	
BD (bn; read as hi)	19,720	2,652	643	
LK (si; read as en)	9,032	0	147	
MK (mk; read as bg, sq)	5,132	256	0	
TL (tet; read as id, pt)	4,962	0	33	
RU (ru; read as uk)	2,070	0	0	
SK (sk; read as cs)	1,926	52	0	
LA (lo; read as th)	333	9	0	
IN (en; read as hi)	323	0	0	
LU (lb; read as de, fr)	6	0	0	
Regions with a language other than the one read				
Wales	0	0	0	0
Highland and Eilean Siar	3,997	348	0	2
Northern Ireland	3,316	0	0	2
Ireland	18,290	2,220	21	19
Catalan-speaking Spain	30,255	93	58	5
Basque Country and Navarre	1,230	3	1	4
Galicia	1,528	0	0	1
Brittany	7,943	244	6	7
Alsace	0	0	0	0
Friesland	0	0	0	0
Finnmark and Troms	3,818	0	2	2
Norrbotten	2,621	0	11	4
Lapland (Finland)	11,124	1,124	11	5

Checks after the eleventh critiques (amendment 15)

Table A65: Odds ratios of flooding that the margin and the upper limit of each interval of R correspond to, and the direct estimates (exploratory; amendments 6 to 14). Each bound belongs to the interval named in its row; a bound is not a test inversion. Each column has one scale. Nominal: the odds ratio injected into the labels. Estimator: the same effect on the scale of the odds ratios of Eq. (2) of the paper, read from the estimator's values on labels with known injected effects (interpolated on the log scale; beyond the largest effect used for this conversion, an injected odds ratio of 3, written as greater than the estimator's value there). No-effect reference: the mean increment of reassigned names with the regional composition rebuilt (amendment 8); bounds with other references are in Tables S15, S16 and A32 of the Supplement. [a] Monte Carlo 2.5th to 97.5th percentiles over the simulated replicates (50 each), which reflect simulation error only. [b] Effect injected on a latent indicator of truly water-related names, drawn with the misreading rates measured against recorded etymologies. [c] Within 500 m, pairs cluster bootstrap over location groups (2,000 replicates); on all ground, cluster-robust interval. [d] Pairs cluster bootstrap (2,000 replicates). [e] Effects on all ground nearest a water-related name, with every model refitted on the injected labels and the decision table of the protocol; names were found useful at injected odds ratios of 1.25 in 0 of 40, 1.50 in 0 of 40 and 2.00 in 40 of 40 replicates (Table S17 of the Supplement).

	All ground nearest the name		Within 500 m of the settlement	
	Nominal	Estimator	Nominal	Estimator
Injected effects (the baseline model's held-out prediction taken as the true probability; base models fixed unless the row says refitted)				
Odds ratio at the margin	2.33	2.94	> 20	—
effect that differs between settlements (mean odds ratio)	2.30	—	—	—
Odds ratio reached by the upper limit of each interval of R (limit in parentheses)				
Protocol's interval (0.003; Monte Carlo range)[a]	1.25 (1.24–1.26)	1.36	6.25 (6.15–6.36)	> 4.73
Spread over fold assignments (amendment 5; 0.004)	1.31	1.45	7.77	> 4.73
Location groups, second stage refitted (amendment 8; 0.013)	1.57	1.83	18.37	> 4.73
Activations × countries, second stage refitted (amendment 11; 0.036)	2.11	2.63	> 20	—
Test on the ratio (a): location groups, models fixed (amendment 13; 0.004)	1.29	1.43	7.40	> 4.73
Test on the ratio (b): location groups, refitted (0.016)	1.65	1.95	> 20	—
Test on the ratio (c): activations within countries, refitted (0.031)	1.99	2.45	> 20	—
Test on the ratio (d): (c) with fold-assignment variance (0.031)	1.99	2.45	> 20	—
for truly water-related names[b]	1.40	—	11.01	—
flips correlated within the ground of each settlement	1.27	1.39	—	—
effect that differs between settlements (mean odds ratio)	1.23	—	—	—
Lower bound for effects that remove flooding	0.70	—	< 0.05	—
Detectable with 80 % power, normal approximation (fixed; refitted)	1.56; 2.00	—	17.80; > 20	—
Detected in at least 80 % of simulations with every model refitted[e]	2.00	—	—	—
Direct estimates: water-related against other names, given the baseline model (95 % interval)				
EMS labels, capped sample[c]	—	1.02 (0.83–1.25)	—	1.01 (0.77–1.31)
EMS labels, every evaluable cell[d]	—	—	—	0.91 (0.76–1.16)
GFM labels, every assessable cell[d]	—	—	—	0.82 (0.60–1.07)
EMS labels on the same cells[d]	—	—	—	0.89 (0.67–1.16)

Table A66: The ground within 500 m of settlements with and without water-related names (exploratory, amendment 15, item B5; capped EMS table, frozen reading): mean shares and their difference within AOIs (AOI fixed effects, 95 % intervals clustered by location group). The rule fixed before computing: if every difference is below 0.05 in absolute value with its interval within ±0.05, the EMS odds ratios near settlements are described as bounds on the association, provided detection errors are the same around both groups. The rule was met.

Quantity	Water-related	Other	Difference (95 %)	Cells
Built-up share (WorldCover 10 m)	0.125	0.139	−0.013 (−0.023 to −0.003)	160,239
Tree share (WorldCover 10 m)	0.303	0.306	−0.002 (−0.013 to 0.008)	160,239
Share of pixels GFM cannot assess (GFM AOIs)	0.638	0.649	−0.030 (−0.045 to −0.015)	103,196
Share of permanent water	0.005	0.004	0.001 (−0.0005 to 0.002)	160,239

Table A67: Inference for the odds ratios near settlements on the cells GFM can assess (exploratory, amendment 15, item B10): cluster-robust 95 % intervals with normal and $t(G-1)$ critical values, an unrestricted wild cluster score bootstrap over location groups (Webb weights, 9,999 draws), and the odds ratio over 200 permutations of the labels within AOIs with the names fixed (median, 2.5th–97.5th percentiles and the share of permutations below the observed odds ratio). The lower block gives the estimate of amendment 7 by thirds of flood magnitude on the capped EMS table, with cluster-robust intervals; AOIs whose cells are all flooded or all dry within the sample drop out of the estimate.

Estimate	OR	CR, z	CR, t	Wild	Labels permuted	Groups
Planned GFM estimate	0.55	0.37–0.82	0.36–0.83	0.37–0.81	0.99 (0.78–1.23; 0.0 %)	80
EMS labels on the same cells	0.79	0.58–1.07	0.58–1.07	0.59–1.05	1.00 (0.78–1.22; 4.0 %)	80
Capped EMS table by thirds of flood magnitude (amendment 7): flooded share of the AOI over its share of ground within 2 m of the drainage						
Lowest third (median magnitude 0.006; 167 AOIs, 7 flooded cells near water-related names)	0.40	0.18–0.91				83
Middle third (median magnitude 0.026; 166 AOIs, 45 flooded cells near water-related names)	0.99	0.68–1.45				98
Highest third (median magnitude 0.101; 167 AOIs, 309 flooded cells near water-related names)	1.07	0.78–1.47				94

Table A68: Census of the exploratory interval estimates of amendments 2 to 15 (amendment 15, item B8; unique, valid estimates; file v76_estimates.csv in the code repository), in three families: flood (estimates on the flood labels or the predictions), site (descriptions of the ground near settlements without labels) and nuisance (coefficients of distance bands and of the baseline prediction, and contrasts between distances): number, number excluding the no-effect value, and numbers with Benjamini–Hochberg (Benjamini and Hochberg, 1995) and Benjamini–Yekutieli (Benjamini and Yekutieli, 2001) q below 0.05 within each family, with the direction of those excluding the no-effect value (favours or against the claim about names).

Family	Estimates	Excluding null	BH $q < 0.05$	BY $q < 0.05$	Direction
flood	866	176	74	57	unclassified 66, against 58, favours 52
nuisance	146	126	126	114	not applicable 126
site	190	98	79	44	favours 66, unclassified 30, against 2
All	1,202	400	279	215	

Checks after the twelfth critiques (amendment 16)

Table A69: One-sided 95 % upper limits of R under every construction with saved resamples, with the rule for few clusters fixed in amendment 16 (item B3): ν is the Satterthwaite effective number of clusters of $L_+(\theta) = \Delta - \theta D$ evaluated at the upper limit θ_U, iterated with the critical value $t_{0.95}(\nu-1)$; "no bound" when $\nu-1 < 1$ or the Fieller interval is unbounded. G: resampled units. The odds ratios are the injected odds ratios on all ground nearest a water-related name that each upper limit reaches, as a range over the four no-effect references of amendment 13 (item B4). Constructions whose limit with $t(\nu-1)$ exceeds the margin or is unbounded: 7.

Construction	R	G	Upper z	Upper $t(\nu)$	Upper $t(\nu-1)$	ν	OR (z)	OR ($t(\nu-1)$)
Protocol's run: location groups, models fixed	−0.003	195	0.003	0.006	0.007	4.3	1.20–1.50	1.31–1.60
(a) Location groups, models fixed	−0.004	195	0.004	0.011	no bound	2.2	1.21–1.51	no bound
(b) Location groups, second stage refitted	−0.004	195	0.016	0.026	0.034	3.4	1.58–1.82	2.01–2.22
(c) Activations within countries, refitted	−0.004	55	0.031	0.064	no bound	2.9	1.93–2.16	no bound
(d) As (c), with the variance of fold assignments	−0.004	55	0.031	0.065	no bound	2.9	1.93–2.16	no bound
Location groups × activations (pigeonhole), refitted	−0.004	188	0.067	no bound	no bound	2.7	2.67–2.83	no bound
Split keeping activations whole: super-clusters, refitted	0.004	106	0.040	0.078	no bound	2.7	2.14–2.32	no bound
the same, names centred within AOIs	−0.003	106	0.002	0.008	no bound	2.5	1.15–1.46	no bound
Designated: super-clusters of the primary split, refitted	−0.003	106	0.014	0.030	0.141	2.5	1.53–1.78	3.97–4.18
the same, names centred within AOIs	−0.004	106	0.001	0.003	0.004	4.6	1.14–1.44	1.20–1.50

Table A70: Where the names increment comes from, and checks of the readings (exploratory, amendment 16, item B9; frozen reading, the designated refit of item B1). Contribution: the group's or block's share of the pooled $\Delta = \mathrm{BS(M1c)} - \mathrm{BS(M3n)}$ (negative: names worsened the predictions); forms: the lexicon's water-related forms matched by the nearest names of the group's cells, with the number of cells. Blocks: the languages the lexicon assigns to the country of the nearest settlement.

Group, block or subset	Share	Contribution	Cells	Forms
Location groups with the largest shares of the variance of Δ				
g00175 (Pakistan)	64 %	-2.6×10^{-5}	10,000; 1.3 % flooded; radar	*chah* (word; 1071), *bela* (word; 85), *dara* (word; 39)
g00127 (Australia)	13 %	1.1×10^{-5}	4,190; 4.1 % flooded; radar	*meadow* (suffix; 257), *meadow* (word; 119), *lower* (word; 94)
g00025 (Bangladesh)	3.0 %	-5.7×10^{-6}	10,000; 23 % flooded; unknown	*char* (word; 871), *bil* (word; 100), *khola* (suffix; 36)
g00126 (Australia)	2.3 %	-5.0×10^{-6}	10,000; 28 % flooded; radar	*burn* (suffix; 179), *flat* (word; 130), *billabong* (word; 116)
g00007 (Germany)	2.0 %	4.3×10^{-6}	10,000; 22 % flooded; unknown	*au* (suffix; 678), *thal* (suffix; 263), *beck* (suffix; 98)
g00132 (Pakistan)	1.7 %	3.9×10^{-6}	10,000; 66 % flooded; optical	*ab* (suffix; 130), *darya* (word; 16), *kallar* (word; 16)
g00008 (Germany)	1.7 %	-4.2×10^{-6}	5,015; 11 % flooded; unknown	*au* (suffix; 638), *nieder* (prefix; 7)
g00191 (Portugal)	1.4 %	-3.9×10^{-6}	10,000; 6.8 % flooded; radar	*vale* (word; 534), *agua* (word; 365), *rio* (word; 233)
g00166 (Germany)	1.0 %	2.9×10^{-6}	10,000; 0.7 % flooded; radar	*thal* (suffix; 236), *au* (suffix; 231), *werder* (suffix; 139)
g00030 (Myanmar)	0.9 %	2.8×10^{-6}	10,000; 11 % flooded; unknown	*chaung* (word; 620), *kan* (word; 231), *kyun* (word; 99)
Language blocks with the largest contributions to Δ				
fa+hi	—	-2.3×10^{-5}	61,219	—
hi	—	-6.7×10^{-6}	20,043	—
mg+fr	—	-4.4×10^{-6}	23,568	—
pt	—	-4.3×10^{-6}	20,062	—
es+qu	—	-1.3×10^{-6}	76,440	—
de	—	-1.1×10^{-6}	83,889	—
Subsets of cells (models fixed; R and 90 % interval)				
Nearest settlement not on whole arc-minutes	—	−0.004 (−0.013 to 0.003)	1,472,604	centred −0.004
Nearest settlement coded PPL	—	−0.004 (−0.014 to 0.004)	1,290,881	centred −0.005
GeoNames record last modified before the event	—	−0.003 (−0.019 to 0.008)	941,820	centred −0.004
The block reads the country's first language	—	0.0003 (−0.004 to 0.006)	1,308,823	centred −0.002
Agreement of the two coders of recorded derivations (Cohen's κ*)*				
water 0.94 ; wet 0.97 ; riverine 0.79 ; crossing 0.97 ; low 0.89 ; high 0.90 ; wetside 0.91	—	—	350	—

References

Baugh C, Colonese J, D'Angelo C, et al (2024) Modelled flood inundation for different return period scenarios at the global scale, version 2.1.2. European Commission, Joint Research Centre (JRC) [data set], URL https://jeodpp.jrc.ec.europa.eu/ftp/jrc-opendata/CEMS-GLOFAS/flood_hazard/

Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. Journal of the Royal Statistical Society: Series B (Methodological) 57:289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

Hawker L, Uhe P, Paulo L, et al (2022) A 30 m global map of elevation with forests and buildings removed. Environmental Research Letters 17:024016. https://doi.org/10.1088/1748-9326/ac4d4f

Holm S (1979) A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6:65–70

Kish L (1965) Survey Sampling. Wiley, New York

Murphy AH (1973) A new vector partition of the probability score. Journal of Applied Meteorology 12:595–600. https://doi.org/10.1175/1520-0450(1973)012<0595:ANVPOT>2.0.CO;2

Nadeau C, Bengio Y (2003) Inference for the generalization error. Machine Learning 52(3):239–281. https://doi.org/10.1023/A:1024068626366

Owen AB (2007) The pigeonhole bootstrap. The Annals of Applied Statistics 1(2):386–411. https://doi.org/10.1214/07-AOAS122

Pekel JF, Cottam A, Gorelick N, et al (2016) High-resolution mapping of global surface water and its long-term changes. Nature 540:418–422. https://doi.org/10.1038/nature20584

Wilson EB (1927) Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association 22:209–212. https://doi.org/10.1080/01621459.1927.10502953